

Kansas Frontier Matrix Unified Implementation, Architecture, and Build Manual

Merged from implementation references/manuals, blueprint/architecture reports, and build plans/companions

Kansas Frontier Matrix technical editorial consolidation

2026-05-11

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1 Front matter [sources: Phase 1 ledger; Phase 2 crosswalk; BLD-GREEN front matter; BLD-COMP front matter; IMPL-PIPE front matter; DOC-DIR §0; DOC-ENC §3; DOC-P18 §3]

1.1 0.1 Title page [sources: BLD-GREEN front matter; BLD-COMP front matter; IMPL-PIPE front matter; DOC-DIR §0]

Title: Kansas Frontier Matrix Unified Implementation, Architecture, and Build Manual.

Version: Unified Draft v0.1, Phase 5 assembly.

Date: 2026-05-11.

Status: CONFIRMED consolidation of supplied PDF doctrine and plans; PROPOSED implementation plan; UNKNOWN current repository implementation maturity.

Supersedes: no governing source by itself. This unified PDF consolidates and cross-references the active supplied corpus. It treats BLD-GREEN v1.1 as the controlling greenfield build sequence, IMPL-PIPE v0.3 as the controlling lifecycle/control-loop manual, DOC-DIR as the controlling directory-placement doctrine, and domain/UI/AI reports as lane or subsystem authorities within their boundaries. [BLD-GREEN front matter; IMPL-PIPE front matter; DOC-DIR §0]

CONFIRMED: the supplied corpus contains document-rich KFM doctrine and implementation plans. UNKNOWN: current mounted-repo topology, route names, schema files, CI workflows, runtime behavior, dashboards, source rights, release artifacts, deployment state, branch protections, and emitted receipts unless separately verified. [BLD-COMP §2.1; IMPL-PIPE §1; DOC-P18 §11]

1.2 0.2 Truth-label legend [sources: user request; BLD-GREEN front matter; IMPL-PIPE §1; BLD-COMP §2.2; DOC-P18 §1.1]

Truth label	Verbatim definition
CONFIRMED	supported in attached docs (cite source ID + section).
PROPOSED	design/recommendation; not verified in a repo.
UNKNOWN	not resolvable from supplied corpus.
NEEDS VERIFICATION	checkable, not yet checked.

Memory is not evidence. Recollection is not fact. DENY, ABSTAIN, and ERROR are finite system outcomes, not truth labels. [BLD-GREEN front matter; IMPL-PIPE §1; UIAI-GAI §§1-4]

1.3 0.3 Source ledger [sources: Phase 1 ledger; DOC-P18 §3; DOC-ENC §3; UIAI-GAI §3A]

The source ledger is a control surface, not a bibliography. Duplicate uploads are carried as lineage but are not counted as independent authority. [IMPL-PIPE §3; DOC-P18 §3]

ID	Filename	Class	Primary scope	Authoritative for	Last-updated marker
BLD-GREEN	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	BUILD	Greenfield build plan	Bootstrap order and phase sequencing	v1.1; updated 2026-05-09
BLD-GREEN-DUP	document_pdf.pdf	BUILD	Duplicate of BLD-GREEN	Duplicate lineage only	v1.1; updated 2026-05-09
BLD-COMP	document_pdf(1).pdf	BUILD	Build Companion	Operational mechanics and hidden dependencies	Draft; no explicit date found
IMPL-REF	#KansasFrontierMatrixImplementationReference.pdf	IMPL	Implementation reference	Source-reported public-repo findings; NEEDS VERIFICATION	No explicit date found
IMPL-REF-DUP	document_pdf(2).pdf	IMPL	Duplicate of IMPL-REF	Duplicate lineage only	No explicit date found
IMPL-PIPE	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	IMPL	Pipeline manual	Lifecycle/control-loop semantics	v0.3; 2026-04-30
IMPL-PIPE-DUP	document_pdf(3).pdf	IMPL	Duplicate of IMPL-PIPE	Duplicate lineage only	v0.3; 2026-04-30

ID	Filename	Class	Primary scope	Authoritative for	Last-updated marker
DOC-DIR	DirectoryRules.pdf	DOCTRINE	Directory placement	Responsibility-root law and schema-home convention	Supersedes prior Directory_Rules.pdf
DOC-ENC	kfm_encyclopedia.pdf	DOCTRINE	Domain and capability encyclopedia	Capability/domain atlas	v0.1; 2026-05-05
DOC-P18	KFM_Pass_18_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	DOCTRINE	Pass 18 synthesis	Idea/category atlas and weak-material signals	2026-05-11 UTC
UIAI-GAI	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	UI-AI	Governed AI	Adapter boundaries, finite outcomes, AI receipts	2026-04-20
UIAI-WHOLE	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	UI-AI	Whole UI + governed AI	UI shell and evidence drawer plan	Generated 2026-04-21
UIAI-MAP	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	UI-AI	MapLibre operating architecture	Renderer boundary, Evidence Drawer, Focus Mode	Generated 2026-04-26
UIAI-MASTER	MasterMapLibreComponents-Functions-Features.pdf	UI-AI	Master MapLibre atlas	Tile/renderer/style/UI/release v1.9; 2026-05-10 atlas	
UIAI-OLLAMA	Ollama&UbuntuInformation.pdf	UI-AI	Ollama and Ubuntu guide	Local/private runtime guidance behind governed API	No explicit date found
DOM-AG	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	DOMAIN	Agriculture	Agriculture source roles, validators, proof/API/UI plan	2026-04-21
DOM-ARCH	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	DOMAIN	Archaeology	Exact-location denial, cultural/steward review	2026-04-21
DOM-AIR	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	DOMAIN	Atmosphere / Air	Air, smoke, climate, EO context discipline	2026-04-21
DOM-FAUNA	KFM_Fauna_Architecture_PDF_Only_Report.pdf	DOMAIN	Fauna	Occurrence evidence and geoprivacy	2026-04-21 14:09 UTC
DOM-FLORA	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	DOMAIN	Flora	Rare-plant controls and flora evidence	2026-04-21T14:13:03Z
DOM-GEOL	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	DOMAIN	Geology / natural resources	Geology/resource distinctions	2026-04-21
DOM-HAB	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	DOMAIN	Habitat	Habitat suitability/connectivity/restoration	2026-04-21
DOM-HF	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	DOMAIN	Habitat + fauna slice	Public-safe occurrence habitat-assignment proof	2026-04-21
DOM-HAZ	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	DOMAIN	Hazards	Not-emergency-alerting hazard lane	2026-04-21 UTC
DOM-HYD	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	DOMAIN	Hydrology	WBD/HUC, USGS Water Data, NHDPlus HR, NFHL proof lane	2026-04-21
DOM-PEOPLE	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	DOMAIN	People/Genealogy/DNA/Land	Living-person/DNA restriction and temporal land assertions	2026-04-21
DOM-ROADS	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	DOMAIN	Roads/Rail/Trade	Transport graph and sensitive route posture	2026-04-21
DOM-SETTLEMENT	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	DOMAIN	Settlements/Infrastructure	Settlement/infrastructure object split	2026-04-21
DOM-SOIL	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	DOMAIN	Soil	SSURGO/SDA and soil-adjacent lanes	2026-04-21
REF-ARCGIS	Earth,Space,andEnvironmentalScienceExplorationswithArcGISProed2.pdf	REF	ArcGIS environmental exercises	GIS/environmental science background	Second edition; ArcGIS Pro 3.0
REF-URBAN	GISinSustainableUrbanPlanningandManagement.pdf	REF	Urban GIS/planning	Planning/resilience indicators background	© 2019
REF-GISPRI-MER	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	REF	GIS/cartographic concepts	Representation/projection/background	© 2008
REF-3DGIS	Archaeological3DGIS.pdf	REF	Archaeological 3D GIS	3D archaeology background	First published 2022
REF-SQL	Advanced-SQL-Concepts.pdf	REF	Advanced SQL	SQL analytics background	Advanced Level
REF-AI	AI_Concepts_Using_Python.pdf	REF	AI/Python	General AI/ML background	First Edition; Dec 2024
REF-DDD	Domain-DrivenDesignReference.pdf	REF	DDD	Bounded context / ubiquitous language background	© 2015
REF-TEMPDB	developing-time-oriented-database-applications-in-sql.pdf	REF	Temporal SQL	Valid/transaction/bitemporal modeling background	© 2000
REF-API	DesigningGreatWebAPIs.pdf	REF	API design	Contract/resource/lifecycle design background	July 2015
REF-SCRAP	instant-web-scraping-with-java.pdf	REF	Web scraping	Source acquisition/error-handling background	Aug 2013

1.4 0.4 Reading rule and scope statement [sources: user request; BLD-GREEN front matter; BLD-COMP reading rule; IMPL-PIPE front matter; DOC-DIR §§1-2]

This unified PDF consolidates, deduplicates, reconciles, and structures supplied KFM PDFs. It does not invent new doctrine and does not claim current repo implementation. Implementation claims remain truth-labeled. Source IDs in section headers identify the source set feeding each section, while bracket tags at paragraph ends identify primary provenance. [DOC-DIR §§1-2; BLD-COMP §2]

When sources conflict, the resolution order is: Directory Rules for path/responsibility-root questions; Greenfield Building Plan for bootstrap order; Pipeline Living Implementation Manual for lifecycle/control-loop semantics; domain blueprints inside their lanes; UI-AI reports for UI shell, Evidence Drawer, Focus Mode, and governed-AI adapter boundaries. Remaining conflicts are shown as Reconciliation Notes and resolved only as PROPOSED. [user request; DOC-DIR §2; BLD-GREEN §§21-23; IMPL-PIPE §§7, 16]

1.5 0.5 Master ToC [sources: user locked outline; BLD-GREEN ToC; IMPL-PIPE ToC; BLD-COMP ToC; DOC-ENC ToC; DOC-P18 ToC]

The locked outline is preserved: Front matter; Part I Doctrine and Boundaries; Part II Architecture; Part III Build Plan; Part IV Operations and Governance; Part V Domain Lanes; Part VI Closing; Appendices. [user request]

2 Part I - Doctrine and Boundaries [sources: BLD-GREEN front matter, §§1-2; BLD-COMP §§1-2; IMPL-PIPE §§1, 4, 7, 17, 21, 24; DOC-DIR §§0-5; DOC-ENC §§2-4; DOC-P18 §§1, 3, 9-13; IMPL-REF Executive Summary; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§1-2; DOM-FLORA §§1-2; DOM-HAZ §§1-2]

2.1 1. Executive determination [sources: BLD-GREEN §1; BLD-COMP §1; IMPL-PIPE §1; DOC-DIR §0; DOC-ENC §2; IMPL-REF Executive Summary]

CONFIRMED doctrine: Kansas Frontier Matrix is a Kansas-first, map-first, time-aware, evidence-first, trust-visible spatial knowledge and publication system whose durable public unit is the inspectable claim, not the map, tile, graph, dashboard, vector index, scene, summary, export, or AI answer. Maps, tiles, graphs, dashboards, indexes, vector stores, scenes, summaries, exports, and AI answers are carriers of evidence; they are never sovereign truth. [BLD-GREEN §1; DOC-ENC §2]

CONFIRMED doctrine / PROPOSED implementation posture: the unified document adopts the governance membrane before feature breadth. Responsibility-root structure, source admission, evidence resolution, policy checks, validators, receipts, release manifests, correction paths, rollback targets, and public-safe proof slices come before live connectors, broad UI expansion, or model-runtime integration. [BLD-GREEN §1; BLD-COMP §1.2]

UNKNOWN current implementation maturity: the supplied corpus contains a real conflict about implementation state. BLD-GREEN assumes an empty starting point with no existing codebase, folder tree, schema, contract, API, data store, UI, workflow, source registry, validator, release artifact, dashboard, runtime, deployment, branch, implementation history, or prior generated source tree. IMPL-REF reports earlier connector-reviewed public-repo surface area across governance, lifecycle, runtime app, schema, policy, and testing trees. BLD-COMP preserves those observations as lineage but warns they are not a fresh current checkout inspection. [BLD-GREEN front matter; IMPL-REF Executive Summary; BLD-COMP §§1, 2.1]

Reconciliation Note - implementation state: CONFIRMED for this manuscript: attached documents support KFM doctrine, build plans, domain-lane blueprints, UI/AI boundaries, and implementation recommendations. UNKNOWN for this manuscript: current target-repo topology, branch state, route names, DTOs, CI workflows, runtime behavior, emitted receipts, dashboards, release artifacts, deployment posture, and actual implementation maturity. PROPOSED resolution: treat BLD-GREEN as controlling the greenfield build sequence, IMPL-REF as lineage requiring fresh verification, and mounted-repo evidence as the only way to upgrade current implementation claims. [BLD-GREEN front matter; IMPL-REF Executive Summary; BLD-COMP §§1, 2.1; IMPL-PIPE §1]

2.2 2. KFM core invariants [sources: BLD-GREEN front matter, §2; IMPL-PIPE §§4, 7; BLD-COMP §§1.2, 3; DOC-DIR §§0, 3; DOC-ENC §4]

CONFIRMED core invariant, preserved verbatim: RAW → WORK/QUARANTINE → PROCESSED → CATALOG/TRIPLET → PUBLISHED. Promotion is a governed state transition, not a file move. [DOC-DIR §0; BLD-GREEN front matter; IMPL-PIPE §1]

CONFIRMED doctrine: every consequential claim resolves EvidenceRef → EvidenceBundle; evidence outranks fluent prose; when evidence cannot be resolved, the system should abstain rather than invent support. This invariant governs API answers, Focus Mode answers, map popups, public layer explanations, catalog entries, and release-supporting claims. [BLD-GREEN §2; BLD-COMP §§1.2, 3; IMPL-PIPE §§7-8]

CONFIRMED doctrine: public clients and ordinary UI surfaces consume governed APIs, released artifacts, catalog records, tile services, EvidenceBundles, release manifests, and safe envelopes only. Public RAW, WORK, QUARANTINE, candidate, direct database, direct internal-store, and direct model-runtime access is denied by default. [BLD-GREEN §2; BLD-COMP §§1.2, 23; IMPL-PIPE §§21, 24]

CONFIRMED doctrine / PROPOSED implementation: promotion requires validation, policy decision, proof or citation closure, release manifest, review state, correction path, rollback target, and receipt trail. BLD-GREEN adds PromotionReceipt, RunReceipt, pre-RAW event governance, DSSE/cosign/Rekor signing, Merkle manifests, OPA/Confest policy-as-code, and catalog closure as proposed implementation patterns; those patterns remain PROPOSED until implemented and verified. [BLD-GREEN §§1-2, 7, 16-21; BLD-COMP §§1.2, 21-22]

CONFIRMED doctrine: AI is interpretive and subordinate. It may summarize resolved evidence, draft bounded

explanations, and assist with candidate deltas, but it cannot decide truth, rights, sensitivity, release, source authority, or review state. The IMPL-PIPE control loop likewise cannot publish or approve itself; it can only propose, validate, compile, request review, and preserve rollback lineage. [BLD-GREEN §2; IMPL-PIPE §§1, 16, 20; BLD-COMP §§1.2, 25]

2.3 3. Directory Rules summary: authority boundaries [sources: DOC-DIR §§0-5, §7.4; BLD-GREEN §§3-4; IMPL-PIPE §§9-11; BLD-COMP §§4-5]

CONFIRMED directory law: “Where a file lives encodes who owns it, what governance it answers to, and what lifecycle it belongs to. Topic does not justify a root folder; responsibility does.” Directory Rules are CONFIRMED as canonical placement rules, while any specific path quoted in them remains PROPOSED until verified against mounted-repo evidence. [DOC-DIR §0]

CONFIRMED conflict rule: Directory Rules wins on path and responsibility-root questions. Accepted ADRs may amend Directory Rules only when explicit and not superseded; per-root README files refine but cannot contradict; domain dossiers and prior reports are lineage/proposed for placement; current repo convention that conflicts with Directory Rules is drift to record, not automatic new canon. [DOC-DIR §§2.1-2.5]

CONFIRMED root-placement doctrine: a root folder appears only when it carries repo-wide responsibility for truth/evidence/release/policy, deployable systems or shared packages, lifecycle data or proof objects, validation/tests/infrastructure/runtime operation, or genuinely cross-domain operation. A domain name such as hydrology, soil, fauna, flora, archaeology, roads, hazards, settlements, atmosphere, agriculture, geology, habitat, transport, or people must not become a root folder merely because it is important. [DOC-DIR §3; BLD-GREEN §3]

If the file’s primary responsibility is...	It belongs under...
Explains something to humans	<code>docs/</code>
Indexes what governs what	<code>control_plane/</code>
Defines object meaning	<code>contracts/</code>
Defines machine shape	<code>schemas/</code>
Decides allow / deny / restrict / abstain	<code>policy/</code>
Proves a rule is enforceable	<code>tests/</code>
Holds sample data for tests	<code>fixtures/</code>
Repo-wide validator, generator, builder, checker	<code>tools/</code>
Deployable application	<code>apps/</code>
Shared library used by multiple deployables	<code>packages/</code>
Source-specific fetcher / admitter	<code>connectors/</code>
Executable pipeline logic	<code>pipelines/</code>
Declarative pipeline configuration	<code>pipeline_specs/</code>
Lifecycle data	<code>data/</code>
Release decision, manifest, rollback, correction	<code>release/</code>
Local runtime adapter / harness	<code>runtime/</code>
Deployment, host, network, exposure posture	<code>infra/</code>
Non-secret config defaults / templates	<code>configs/</code>
Database / schema / graph migration	<code>migrations/</code>
Worked runnable example	<code>examples/</code>

CONFIRMED lifecycle placement rule: under `data/`, the lifecycle phase must be explicit: raw, work, quarantine, processed, catalog, triplets, published, receipts, proofs, registry, rollback. Receipts, proofs, registry, and rollback are emitted alongside lifecycle directories and do not replace RAW, WORK/QUARANTINE, PROCESSED, CATALOG/TRIPLET, or PUBLISHED. [DOC-DIR §4]

2.4 4. Truth posture and evidence rule [sources: BLD-GREEN front matter, §2; IMPL-PIPE §§1, 3-4; BLD-COMP §§2-3; DOC-P18 §§1, 3; UIAI-GAI §§1-3; DOC-ENC §§2-4]

CONFIRMED truth posture: this unified PDF uses four truth labels for implementation-state claims: CONFIRMED, PROPOSED, UNKNOWN, and NEEDS VERIFICATION. DENY, ABSTAIN, and ERROR remain system outcomes,

not truth labels. [BLD-GREEN front matter; IMPL-PIPE §1; BLD-COMP §2.2; DOC-P18 §1.1]

CONFIRMED evidence rule: the source ledger is a control surface, not a bibliography. Each source is treated according to what it can support and what it cannot prove. Current mounted repo evidence would outrank documents for implementation behavior, but absent mounted repo evidence keeps runtime, route, workflow, artifact, and deployment claims UNKNOWN. [IMPL-PIPE §3; BLD-COMP §§2.1-2.3; DOC-P18 §3]

CONFIRMED cite-or-abstain rule: a statement needing evidence either cites support or abstains. In KFM terms, an EvidenceRef must resolve to an EvidenceBundle before a consequential claim can be presented as supported; missing, denied, conflicted, stale, or policy-blocked support produces ABSTAIN, DENY, ERROR, REVIEW_NEEDED, quarantine, or a visible limitation rather than confident prose. [BLD-GREEN §2; IMPL-PIPE §§7-8, 17; BLD-COMP §§3, 22]

CONFIRMED implementation-claim rule for this unified PDF: any claim that “the system does X,” “the repo contains Y,” “the route returns Z,” “the validator enforces Q,” or “the UI displays R” must be backed by current implementation evidence or marked PROPOSED, UNKNOWN, or NEEDS VERIFICATION. Attached architecture PDFs can support doctrine and design, but not current deployed behavior unless paired with repo, test, log, workflow, dashboard, emitted artifact, or release evidence. [BLD-COMP §§1, 2.1, 24; IMPL-PIPE §1; DOC-P18 §§1.2, 11]

2.5 5. Publication, rights, and sensitivity posture [sources: BLD-GREEN §§8, 18-20, 26; IMPL-PIPE §§17, 21, 24; BLD-COMP §§8-11, 21-23, 31; DOC-DIR §§0, 3; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§1-2; DOM-FLORA §§1-2; DOM-HAZ §§1-2]

CONFIRMED publication posture: publication is a governed state transition, not a file move. A public or semi-public release requires evidence closure, rights posture, sensitivity posture, validation, policy decision, review state, release manifest, proof pack or equivalent support, correction path, rollback target, stale-state handling, and receipt trail. [BLD-GREEN §§19-20; IMPL-PIPE §21; BLD-COMP §§1.2, 21-22]

CONFIRMED rights posture: unknown rights, unresolved source terms, unclear attribution duties, unknown source role, prohibited source use, or missing SourceActivationDecision blocks public release by default. The safe state is quarantine, denial, restriction, or abstention until rights, source role, access conditions, cadence, and release class are recorded. [BLD-GREEN §§8, 18-19; BLD-COMP §§1.2, 8, 31; IMPL-PIPE §§13, 17]

CONFIRMED sensitivity posture: public-safe output must be downstream of sensitivity review and documented transformation when precise exposure creates risk. Sensitive exact locations, rare species, archaeological sites, cultural or sovereign material, living-person data, DNA/genomic material, infrastructure details, and source-controlled steward information require fail-closed handling, generalization, redaction, staged access, delayed publication, or denial unless release support is explicit. [DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§1-2; DOM-FLORA §§1-2; BLD-COMP §§21-23]

CONFIRMED domain-risk boundary: hazards analysis must not become an emergency alerting system; archaeology must deny exact public site locations by default; people/genealogy/DNA/land must keep living-person and DNA-derived outputs denied or restricted by default; fauna and flora must apply geoprivacy and rare-species controls; infrastructure exposure must be policy-gated. These rules remain lane-specific gates expanded in Part V rather than repeated in every architecture section. [DOM-HAZ §§1-2; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§1-2; DOM-FLORA §§1-2; DOM-SETTLE §§2-4]

3 Part II - Architecture [sources: BLD-GREEN §§1, 5-15, 21; BLD-COMP §§3-8, 10, 21-25; IMPL-PIPE §§6-20; DOC-DIR §§0-5, §7.4; DOC-ENC §§3-8; DOC-P18 §§8.3-8.13, 9-13; UIAI-GAI §§1-4; UIAI-WHOLE §§7-19, 25-28; UIAI-MAP §§1, 4-15; UIAI-MASTER §§2, 9-14; IMPL-REF Executive Summary and Connector Inventory; all DOM-* executive / architecture sections]

3.1 6. System overview [sources: BLD-GREEN §§1, 7-15; BLD-COMP §§3-4; IMPL-PIPE §§6, 15-20; UIAI-MAP §§1, 4-6; UIAI-WHOLE §§7-9; UIAI-GAI §§1-4; DOC-ENC §§2-8]

CONFIRMED doctrine / PROPOSED target architecture: KFM is structured as a governed spatial-evidence system, not as a map app with optional citations. The architectural center is the trust spine: source admission, lifecycle storage, evidence resolution, validation, policy decision, catalog/proof closure, release, governed API, trust-visible UI, and bounded AI interpretation. BLD-GREEN frames this as a greenfield build of the governance membrane before live connectors, broad UI, or AI runtime integration; BLD-COMP reframes it operationally as the trust-object machine that must work before features feel real. [BLD-GREEN §§1, 21; BLD-COMP §§1.2, 3]

CONFIRMED doctrine / UNKNOWN current implementation depth: the supplied corpus supports this architecture as doctrine and plan, but it does not prove current repo behavior. IMPL-PIPE and multiple subsystem reports state that no mounted repo was visible in their runs, while IMPL-REF separately reports earlier public-repo surface area. The unified architecture treats route names, DTOs, package manager, schema files, validators, CI workflows, emitted receipts, runtime logs, dashboards, release artifacts, deployment posture, and UI component homes as PROPOSED until mounted-repo evidence verifies them. [IMPL-PIPE §1; IMPL-REF Executive Summary; BLD-COMP §2.1]

PROPOSED architectural cut: the smallest sound KFM system has five cooperating planes: governance/control plane, lifecycle/data plane, evidence/catalog/proof plane, governed service/API plane, and user interaction plane. AI, MapLibre, graph projections, vector indexes, PMTiles, scenes, summaries, and dashboards are downstream carriers; none replaces source authority, EvidenceBundle resolution, PolicyDecision, PromotionDecision, release manifest, or rollback target. [BLD-GREEN §§7-15; UIAI-MAP §§4-6; UIAI-GAI §§1-4; DOC-ENC §§4-8]

3.2 7. Canonical data lifecycle [sources: BLD-GREEN front matter, §§5, 9, 21; IMPL-PIPE §7; BLD-COMP §§3, 9; DOC-DIR §§0, 3-4; domain lane reports]

CONFIRMED lifecycle doctrine, preserved as the governing chain: RAW → WORK/QUARANTINE → PROCESSED → CATALOG/TRIPLET → PUBLISHED. Promotion is a governed state transition, not a file move. BLD-GREEN v1.1 adds a pre-RAW event family for the admission edge, but that addition does not change the core invariant; it governs what happens before material is accepted into RAW. [BLD-GREEN front matter; DOC-DIR §0; IMPL-PIPE §7]

CONFIRMED lifecycle semantics: RAW preserves admitted source material under source identity; WORK hosts transformations and unresolved candidates; QUARANTINE holds rights, sensitivity, validation, source-role, temporal, or evidence defects; PROCESSED contains normalized outputs that have passed transformation checks; CATALOG/TRIPLET records claim, layer, graph, provenance, and discovery surfaces; PUBLISHED exposes only released, policy-allowed, reviewable, rollback-capable artifacts. [IMPL-PIPE §7; BLD-COMP §§3, 9; DOC-DIR §§3-4]

PROPOSED pre-RAW admission edge: `event_envelope`, `prefilter_output`, and `event_run_receipt` should record attempted intake before data is admitted into RAW, especially where automated watchers, GitOps PR emission, live feeds, source refreshes, or model-assisted candidate generation could otherwise blur the boundary between observed input and accepted source material. [BLD-GREEN §§1, 7, 21; BLD-COMP §§8-9]

3.3 8. Canonical object families [sources: BLD-GREEN §§7, 10-11; IMPL-PIPE §§8, 12, 16, 21; BLD-COMP §§3, 6, 8, 10; UIAI-GAI §§2-4; DOC-P18 §§8.5, 8.10; UIAI-MASTER §§10-12]

CONFIRMED doctrine / PROPOSED implementation plan: KFM's object families make trust inspectable. Each family carries a specific governance burden such as source identity, evidence closure, policy decision, validation result,

release state, correction lineage, or rollback target. [BLD-GREEN §§7, 10-11; IMPL-PIPE §8; BLD-COMP §§3, 6]

Object family	Architectural role	Status and source
EvidenceRef	Pointer from claim/feature/answer/layer/proof item to evidence support	CONFIRMED doctrine [BLD-GREEN §10; IMPL-PIPE §8]
EvidenceBundle	Resolved evidence package with source, provenance, scope, citation, and review context	CONFIRMED doctrine [BLD-GREEN §10; UIAI-GAI §§2-4]
SourceDescriptor	Source identity, role, rights, cadence, access, sensitivity, release posture	CONFIRMED doctrine / PROPOSED implementation [BLD-GREEN §9; BLD-COMP §8]
SourceActivationDecision	Gate deciding source use, restriction, quarantine, or denial	PROPOSED implementation [BLD-COMP §8; IMPL-PIPE §13]
PolicyDecision / DecisionEnvelope	Finite allow/deny/restrict/abstain/error decision envelope	CONFIRMED doctrine / PROPOSED implementation [UIAI-GAI §§2-4; DOC-P18 §8.6]
RunReceipt	Auditable record of intake, transform, validation, catalog, release, or rebuild action	CONFIRMED doctrine / PROPOSED implementation [BLD-GREEN §7; DOC-P18 §8.5]
PromotionReceipt	Auditable representation of Promotion Gates A-G	CONFIRMED doctrine / PROPOSED implementation [BLD-GREEN §§1, 7]
ReleaseManifest / MapReleaseManifest	Published artifact set, digests, policy posture, rollback target	CONFIRMED doctrine / PROPOSED implementation [BLD-GREEN §§19-20; UIAI-MASTER §§10-12]
AIReceipt / RuntimeResponseEnvelope	Bounded AI request/response accountability with evidence, citations, outcome	PROPOSED implementation [UIAI-GAI §§2-4; UIAI-OLLAMA §§9-13]
QueryRunRecord / CandidateDelta / RecompileManifest	Loop-control objects for query-save-validate-compile-review-promote-recompile	PROPOSED implementation [IMPL-PIPE §16]

CONFIRMED anti-collapse rule: catalogs, triplets, graph projections, PMTiles, layer manifests, model outputs, summaries, and UI answers are derivative or publication surfaces; they do not become root truth. Their claims must remain traceable back through EvidenceRef, EvidenceBundle, receipts, policy decisions, and release records. [BLD-GREEN §§1, 10-11; IMPL-PIPE §§8, 21; UIAI-MASTER §§2, 10-12]

3.4 9. Repository architecture plan [sources: DOC-DIR §§0-5, §7.4; BLD-GREEN §§3-4; IMPL-PIPE §§9-10; BLD-COMP §§4-5; IMPL-REF Connector Inventory]

CONFIRMED directory doctrine: repository architecture is responsibility-rooted. A file's location encodes ownership, governance, lifecycle, and authority; topic alone does not justify a root folder. Directory Rules therefore governs placement before domain blueprints, prior scaffold reports, or implementation-reference path proposals. [DOC-DIR §§0-3]

CONFIRMED responsibility-root plan: canonical root families include human explanation, control-plane indexing, contracts, schemas, policy, tests, fixtures, tools, scripts, apps, packages, connectors, pipelines, pipeline specs, lifecycle data, release, runtime, infrastructure, configs, migrations, and examples. Domains live under these roots as lanes, not as new root folders. [DOC-DIR §§3-4; BLD-GREEN §§3-4]

Reconciliation Note - greenfield plan versus earlier public-repo report: IMPL-REF reports an earlier public monorepo surface including `.github`, `apps`, `configs`, `connectors`, `contracts`, `control_plane`, `data`, `docs`, `fixtures`, `infra`, `jsonschema`, `migrations`, `packages`, `pipeline_specs`, `pipelines`, `policies`, `policy`, `release`, `runtime`, `schemas`, `scripts`, `styles`, `tests`, `tools`, `ui`, `viewer_templates`, and `web`; BLD-GREEN assumes no

existing repo. PROPOSED resolution: use BLD-GREEN for bootstrap order, DOC-DIR for placement law, and treat IMPL-REF as NEEDS VERIFICATION lineage until the current repo is mounted and inspected. [IMPL-REF Connector Inventory; BLD-GREEN front matter; DOC-DIR §2.5]

3.5 10. Schema and contract plan [sources: DOC-DIR §§2.4, 7.4; IMPL-PIPE §11; BLD-COMP §5; BLD-GREEN §§7, 16, 21; IMPL-REF schema-home discussion; domain schema sections]

CONFIRMED directory rule / PROPOSED implementation posture: Directory Rules states a schema-home convention of `schemas/contracts/v1/<...>` as the default per ADR-0001, and it requires an ADR before changing the schema-home rule or creating parallel schema, contract, policy, source, registry, release, proof, or receipt homes. Specific schema paths remain PROPOSED until mounted-repo evidence verifies them. [DOC-DIR §§0, 2.4, 7.4]

CONFIRMED split of responsibilities: `contracts/` defines object meaning, service commitments, and domain vocabulary; `schemas/` defines machine shape and validation; `policy/` decides allow, deny, restrict, or abstain; `tests/` proves enforceability; `fixtures/` supplies valid and invalid sample cases. [DOC-DIR §4; BLD-COMP §5; IMPL-PIPE §§11, 17, 22]

Reconciliation Note - contracts versus schemas: IMPL-REF reports earlier ambiguity between `contracts/` and `schemas/`, while DOC-DIR now supplies a default schema-home convention and ADR discipline. PROPOSED resolution: use DOC-DIR as governing placement doctrine; preserve any mounted-repo `contracts/` or `jsonschema/` conventions as compatibility roots only after inspection; require an ADR or migration note before creating or moving authoritative schema homes. [IMPL-REF Executive Summary; DOC-DIR §§2.4, 7.4; BLD-COMP §5.3]

3.6 11. Source registry architecture [sources: BLD-GREEN §9; IMPL-PIPE §13; BLD-COMP §8; DOC-P18 §8.3; DOC-ENC §3; domain source-registry sections]

CONFIRMED doctrine: the source registry is an admission and authority-control surface, not a bibliography. It records source identity, role, rights posture, access method, cadence, steward, sensitivity, freshness expectations, attribution requirements, and public-release class so source material can be admitted, quarantined, restricted, or denied before it shapes public claims. [BLD-GREEN §9; IMPL-PIPE §13; BLD-COMP §8]

PROPOSED source activation flow: create or update SourceDescriptor; review source role, rights, sensitivity, cadence, and access; issue SourceActivationDecision declaring allowed, restricted, denied, or needs-review use; keep connectors/watchers inactive until activation decision, fixtures, validators, and policy gates exist. [BLD-COMP §§8.1-8.2; IMPL-PIPE §13; DOC-P18 §8.3]

CONFIRMED cross-domain rule: source role cannot be inferred from convenience. A community-science occurrence source is not a legal-status authority; assessor rows are not title truth; regulatory flood layers are not observed inundation; operational warnings are not life-safety authority inside KFM; remote-sensing anomalies are candidates until reviewed. [DOM-FAUNA §§1-2; DOM-PEOPLE §§2-4; DOM-HYD §§2-4; DOM-HAZ §§1-2; DOM-ARCH §§2-6]

3.7 12. Domain lane architecture [sources: BLD-GREEN §6; IMPL-PIPE §14; DOC-ENC §§5, 7; all DOM-* reports; DOM-HF]

CONFIRMED architecture pattern: domain lanes are bounded responsibility areas inside shared KFM governance. They may define domain object families, source roles, validators, policy gates, pipelines, catalog entries, graph edges, UI layers, Evidence Drawer payloads, and Focus Mode constraints, but they do not own root-folder authority, global schema-home decisions, publication law, or public bypasses around governed APIs. [BLD-GREEN §6; IMPL-PIPE §14; DOC-DIR §§3-4]

Lane	Primary architectural emphasis	Governing source IDs
Hydrology	Early proof lane; WBD/HUC, USGS Water Data, NHDPlus HR, NFHL, catalog/proof closure	DOM-HYD; BLD-GREEN §21; IMPL-PIPE §14
Soil	SSURGO/SDA, soil properties, soil moisture, agriculture/hydrology adjacency	DOM-SOIL; DOM-AG; DOM-HYD

Lane	Primary architectural emphasis	Governing source IDs
Habitat	Habitat patches, suitability, connectivity, restoration, public-safe products	DOM-HAB; DOM-HF; DOM-FAUNA; DOM-FLORA
Fauna	Taxonomy, occurrence evidence, geoprivacy, range/habitat relations	DOM-FAUNA; DOM-HF
Flora	Plant taxonomy, specimens, rare-plant controls, invasives, phenology	DOM-FLORA; DOM-HAB; DOM-FAUNA
Hazards	Analysis/history/regulatory/operational context; not emergency alerting	DOM-HAZ; DOM-HYD; DOM-AIR
Archaeology	Cultural sensitivity, exact-location denial, remote-sensing humility, steward review	DOM-ARCH; REF-3DGIS
Roads/Rail/Trade	Transport graph, routes, corridors, facilities, sensitive cultural-route handling	DOM-ROADS; DOM-SETTLE; DOM-ARCH
Settlements/Infrastructure	Settlement/infrastructure object separation, sensitive infrastructure controls	DOM-SETTLE; DOM-ROADS
Atmosphere/Air	Observations, AQI, climate, smoke, EO context, knowledge-character labeling	DOM-AIR; DOM-HAZ
Geology/Natural Resources	Geology/resource distinction, boreholes, public-safe geometry	DOM-GEOL; DOM-SOIL; DOM-HYD
Agriculture	Crops, fields, management, soil/water adjacency, source-role controls	DOM-AG; DOM-SOIL; DOM-HYD
People/Genealogy/DNA/Land	Assertion-first genealogy, DNA restriction, temporal land-ownership assertions	DOM-PEOPLE; REF-TEMPDB; REF-DDD

PROPOSED lane dependency order: early implementation should favor low-sensitivity, proof-bearing, public-safe slices such as hydrology or habitat/fauna synthetic fixtures before lanes with living-person, DNA, exact archaeology, critical infrastructure, or rare-species exposure risk. [BLD-GREEN §§6, 21-22; IMPL-PIPE §§14, 26; DOM-HYD §§2-4; DOM-HF §§1-5]

3.8 13. API architecture [sources: BLD-GREEN §12; IMPL-PIPE §18; UIAI-WHOLE §17; UIAI-MAP §7; DOC-P18 §8.11; REF-API]

CONFIRMED doctrine / PROPOSED API architecture: KFM APIs are governed interfaces over released or reviewable trust objects, not raw database windows. Public API payloads should expose finite outcome states, evidence references, policy posture, temporal and spatial scope, stale state, release state, correction lineage, and identifiers that allow EvidenceBundle inspection. [BLD-GREEN §12; IMPL-PIPE §18; UIAI-MAP §7; DOC-P18 §8.11]

PROPOSED API design discipline: API contracts should be design-first, resource-oriented, lifecycle-aware, and documented before UI or model-runtime coupling. REF-API supports contract-first resource and HTTP lifecycle design as background, while KFM doctrine adds stricter requirements: no public RAW/WORK/QUARANTINE routes, no direct internal-store handles, no direct model-client path, and no uncited consequential answer. [REF-API Chs. 1-4; BLD-GREEN §12; IMPL-PIPE §18; UIAI-GAI §§2-4]

UNKNOWN implementation details: actual API framework, route naming, OpenAPI location, DTO names, authentication middleware, authorization model, graph access layer, tile-service mediation, and review-console endpoints are not established by the supplied corpus. Any route examples in source PDFs are PROPOSED placeholders until a current checkout is inspected. [IMPL-PIPE §1; UIAI-WHOLE §§1, 17; BLD-COMP §2.1]

3.9 14. UI, MapLibre, Evidence Drawer, and Focus Mode [sources: UIAI-MAP §§1, 4-12; UIAI-WHOLE §§7-19; UIAI-MASTER §§2, 9-13; BLD-GREEN §§13-14; IMPL-PIPE §19; DOC-P18 §§8.12-8.13]

CONFIRMED UI doctrine: MapLibre is the disciplined 2D renderer and interaction runtime inside the governed KFM shell; it is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority, or AI authority. UIAI-MAP states the one-sentence rule as: the renderer is downstream of trust, never upstream of it. [UIAI-MAP §§1, 4-5]

PROPOSED UI architecture: the governed shell should expose map state, time state, layer release state, stale state, evidence state, policy state, review state, correction lineage, and negative states at the point of use. The core interaction slice is: released layer -> selected feature candidate -> governed API -> EvidenceBundle -> Evidence Drawer -> bounded Focus Mode answer, abstention, denial, or error. [UIAI-MAP §§6-12; UIAI-WHOLE §§18-19; IMPL-PIPE §19]

CONFIRMED / PROPOSED artifact boundary: tiles, PMTiles, MVT, MLT, COGs, style JSON, screenshots, Story Nodes, 3D scenes, graph projections, catalog records, and AI answers are downstream carriers and must remain tied to release manifests, digests, EvidenceBundles, policy decisions, and rollback targets. UIAI-MASTER v1.9 strengthens this with PMTiles/COG/GeoParquet provenance, artifact digest, STAC/DCAT/PROV, OCI/ORAS, Cosign, and verification backlog patterns; those patterns remain PROPOSED unless implemented and verified. [UIAI-MASTER §§2, 9-14; UIAI-MAP §§10-14]

NEEDS VERIFICATION technical fact: MapLibre package versions, plugin status, MLT readiness, hosting headers, PMTiles tooling, browser/device performance, mobile/native parity, dependency allowlists, and production rendering behavior are version- and environment-sensitive. The unified architecture records the MapLibre operating law and object boundaries, not a package pin or implementation claim. [UIAI-MAP §§1-2, 11, 14-15; UIAI-MASTER §§2, 14]

3.10 15. Governed AI architecture [sources: UIAI-GAI §§1-4; UIAI-OLLAMA §§1-16; UIAI-WHOLE §§18-25; IMPL-PIPE §20; BLD-GREEN §15; DOC-P18 §§8.8, 8.13]

CONFIRMED AI doctrine: AI is interpretive, not authoritative. It may synthesize over admissible evidence, draft bounded explanations, support retrieval acceleration, or help evaluate candidate deltas, but EvidenceBundle, source authority, policy decision, review state, release state, and citation validation outrank generated language. UIAI-GAI recommends beginning with a provider-neutral, evidence-subordinate runtime slice using MockAdapter, policy precheck, citation validation, finite RuntimeResponseEnvelope, AIRceipt, RunReceipt, and source-coverage checks. [UIAI-GAI §§1-4; IMPL-PIPE §20]

PROPOSED AI request flow: define scope; resolve EvidenceRef to EvidenceBundle; apply rights, sensitivity, role, and release checks; provide only admissible context to the model adapter; require structured output; validate citations and policy after generation; emit ANSWER, ABSTAIN, DENY, or ERROR with receipt and inspectable support. [UIAI-GAI §§2-4; UIAI-OLLAMA §§5-13; UIAI-WHOLE §§18-25]

Reconciliation Note - provider-neutral AI versus Ollama guidance: UIAI-GAI says the first implementation slice should not start with Ollama, OpenAI, a browser chat panel, or UI polish; UIAI-OLLAMA explains how Ollama can be used as a replaceable local or privately hosted model runtime behind the governed API. PROPOSED resolution: fix adapter contracts, evidence gates, finite envelopes, citation validation, and receipts first; only then admit Ollama or another provider behind the same governed interface. [UIAI-GAI §2; UIAI-OLLAMA §§1, 4-11]

CONFIRMED failure boundary: model output must not read RAW, WORK, QUARANTINE, unpublished candidate data, direct canonical stores, or sensitive exact-location material unless a governed, policy-allowed internal workflow explicitly permits bounded review use. Public model answers must not bypass evidence resolution, citation validation, policy gates, release state, correction lineage, or rollback discipline. [UIAI-OLLAMA §§1-3, 9-15; UIAI-GAI §§2-4; BLD-GREEN §15]

4 Part III - Build Plan [sources: BLD-GREEN §§1-5, 16-28; BLD-COMP §§1, 3-5, 20-31, Appendix A-C; IMPL-PIPE §§1-3, 22-28; DOC-DIR §§0-5, §7.4; IMPL-REF Executive Summary and Implementation Plan]

4.1 16. Greenfield bootstrap principles [sources: BLD-GREEN front matter, §§1-2, 21-22; BLD-COMP §§1.2, 3-4; IMPL-PIPE §§1, 25-26; IMPL-REF Executive Summary]

CONFIRMED doctrine / PROPOSED implementation: the controlling bootstrap posture is “governance membrane first.” BLD-GREEN v1.1 assumes an empty starting point and proposes creating responsibility-root skeleton, doctrine docs, control-plane registers, source descriptors, pre-RAW event schemas, core schemas, no-network fixtures, validators, policy stubs, finite envelopes, release dry-run, correction/rollback objects, and one public-safe proof slice before live connectors, broad UI expansion, or AI runtime integration. [BLD-GREEN front matter, §§1, 21]

CONFIRMED reconciliation rule: BLD-GREEN wins greenfield bootstrap order, but IMPL-REF reports an earlier connector-reviewed public repository with meaningful implementation surface. PROPOSED resolution: use BLD-GREEN as normative greenfield sequence, preserve IMPL-REF as NEEDS VERIFICATION lineage, and upgrade no current implementation claim without a fresh mounted-repo inspection. [BLD-GREEN front matter; IMPL-REF Executive Summary; BLD-COMP §§1, 2.1]

PROPOSED bootstrap dependency order: do not start with UI, live connectors, or AI. BLD-COMP states the safer order as meaning -> shape -> examples -> validators -> policy -> receipts -> resolver -> release -> UI -> AI, because each public-facing claim must earn its way outward. [BLD-COMP §4]

PROPOSED minimum first-release thesis: the first release should prove one small trust path, not broad feature coverage. The accepted first proof pattern is a fixture-only or no-network public-safe slice that exercises source activation, EvidenceRef -> EvidenceBundle closure, policy denial, release dry-run, governed API envelope, Evidence Drawer payload, Focus Mode MockAdapter behavior, correction, and rollback. [BLD-COMP Final build recommendation; BLD-GREEN §21; IMPL-PIPE §26]

4.2 17. Directory-by-directory decisions [sources: DOC-DIR §§0-5, §7.4; BLD-GREEN §§3-5, §21; BLD-COMP §§4-5; IMPL-PIPE §§9-11]

CONFIRMED directory basis: Directory Rules governs path and responsibility-root questions. A specific path quoted in any build plan remains PROPOSED until checked against mounted-repo evidence, and any conflict between mounted repo convention and Directory Rules must be recorded as drift or resolved by ADR rather than silently treated as canon. [DOC-DIR §§0, 2.4-2.5]

CONFIRMED / PROPOSED root plan: BLD-GREEN v1.1 keeps the responsibility-root monorepo baseline and adds no new root folders for v1.1 features. It places pre-RAW event schemas under `schemas/contracts/v1/events/`, watcher specs under `pipeline_specs/watchers/`, OPA bundles under `policy/opa/`, attestation tooling under `tools/attest/`, promotion-gate validators under `tools/validators/promotion_gate/`, and pre-RAW transient event staging under PROPOSED `data/events/`, which is DENY for public access. [BLD-GREEN §§3-5; DOC-DIR §§3-4]

Decision area	Unified decision	Status
Root folders	Use responsibility roots, not domain roots.	CONFIRMED doctrine
Domain lanes	Place domain material under owning roots.	CONFIRMED doctrine
Schema home	Default to <code>schemas/contracts/v1/<...></code> unless ADR changes it.	CONFIRMED doctrine / PROPOSED paths
Policy bundles	Route OPA / Conftest bundles through <code>policy/opa/</code> .	PROPOSED-to-create
Pre-RAW events	Use event schemas and <code>data/events/</code> transient staging.	PROPOSED-to-create

Decision area	Unified decision	Status
Watchers	Place watcher specs under <code>pipeline_specs/watchers/</code> ; no direct main commits.	PROPOSED-to-create
Attestation	Place signing helpers under <code>tools/attest/</code> .	PROPOSED-to-create
Promotion checks	Place gate validators under <code>validator</code> roots, not release roots.	PROPOSED-to-create

CONFIRMED compatibility warning: IMPL-REF lists earlier public-repo roots such as `jsonschema`, `policies`, `styles`, `ui`, `viewer_templates`, and `web`; these are not promoted to current authority by this unified build plan. PROPOSED resolution: treat them as compatibility or lineage roots until current repo is mounted, then record an ADR or drift entry before merging, removing, or elevating any parallel schema, policy, UI, or proof home. [IMPL-REF Connector Inventory; DOC-DIR §§2.4-2.5; BLD-COMP §5.3]

4.3 18. Initial implementation package [sources: BLD-GREEN §21; BLD-COMP §§3-5, 20, Appendix C; IMPL-PIPE §26]

PROPOSED initial package: the first implementation package should be no-network and fixture-driven. BLD-GREEN v1.1 updates the package to include event, receipt, promotion, Merkle, STAC-profile, OPA, catalog-integrity, and replay components; BLD-COMP narrows the practical next proof to a hydrology fixture PR with SourceDescriptor, EvidenceRef, EvidenceBundle, PolicyDecision, RuntimeResponseEnvelope, ReleaseManifest, RollbackCard, valid/invalid fixtures, validator commands, and one Evidence Drawer payload. [BLD-GREEN §21; BLD-COMP Appendix C and Final build recommendation]

Package element	Required content	Status
Source/evidence objects	SourceDescriptor, EvidenceRef, EvidenceBundle	PROPOSED
Decision/runtime objects	PolicyDecision, RuntimeResponseEnvelope	PROPOSED
Release/rollback objects	ReleaseManifest, RollbackCard, correction linkage	PROPOSED
v1.1 event family	event_envelope, prefilter_output, event_run_receipt	PROPOSED
Receipt/promotion family	RunReceipt, DSSE envelope, PromotionReceipt	PROPOSED
Integrity family	MerkleManifest, spec_hash, signed receipt checks	PROPOSED
Catalog closure	KFM STAC profile, PROV/STAC/DCAT emitters, integrity validator	PROPOSED
Tests and fixtures	Valid/invalid no-network fixtures and deny/abstain/error cases	PROPOSED
Policy bootstrap	OPA packages for access, obligations, gates, release, STAC	PROPOSED

PROPOSED first six PRs: BLD-COMP recommends a deliberately boring opening sequence: PR-001 path and authority baseline; PR-002 core object semantics; PR-003 schema and fixture wave; PR-004 policy negative suite; PR-005 evidence closure dry-run; and a sixth release dry-run / receipt-oriented proof slice before public UI or connector work. [BLD-COMP §4.2]

PROPOSED control-loop increment: IMPL-PIPE defines the next smallest safe PR as control-loop contracts, fixtures, validators, and a dry-run compiler plan, explicitly excluding live connectors, public UI routes, model runtime adapters, PUBLISHED artifacts, production source registries, and sensitive domain data. [IMPL-PIPE §26]

4.4 19. Build phases and milestones [sources: BLD-GREEN §§22-23; BLD-COMP §4; IMPL-PIPE §25; IMPL-REF Implementation Plan]

CONFIRMED Greenfield sequencing boundary: BLD-GREEN v1.1 states that phases 22.0-22.15 from v1.0 are retained and adds two named phases in v1.1: “Phase 16: Signing and Attestation Pipeline” and “Phase 17: Pre-RAW Event Family, GitOps Watchers, Catalog Closure.” The supplied v1.1 text does not restate the full names of retained phases 22.0-22.15, so this unified document does not invent those phase names. [BLD-GREEN §22]

Greenfield v1.1 added phase	Goal	Done criterion
Phase 16: Signing and Attestation Pipeline	Implement RunReceipt + DSSE signing and PromotionReceipt for all lifecycle stages.	Every lifecycle stage emits a RunReceipt; every promotion produces a PromotionReceipt with all seven gates.
Phase 17: Pre-RAW Event Family, GitOps Watchers, Catalog Closure	Implement pre-RAW events, GitOps watchers, Merkle manifests, OPA bootstrap, KFM STAC profile, and catalog emitters.	Watcher detects material change -> event_run_receipt -> PR -> gates -> PromotionReceipt -> Merkle manifest -> PROV/STAC/DCAT closure.

CONFIRMED milestone additions: BLD-GREEN v1.1 retains milestones M0-M9 from v1.0 and adds M10 “Signing baseline” and M11 “Full trust membrane.” M10 requires lifecycle RunReceipts, Promotion Gates A-G passing for a fixture slice, and a produced/verified PromotionReceipt; M11 requires the pre-RAW layer, watcher PR-emission, Merkle proof-pack field, OPA bootstrap, KFM STAC enforcement, and catalog closure emitters producing PROV + STAC + DCAT with integrity validation. [BLD-GREEN §23]

PROPOSED operational phase overlay: IMPL-PIPE provides the current loop-oriented roadmap from repo inspection through domain expansion. It begins with Phase 0 repo inspection and evidence inventory; then documentation control plane; shared schemas and fixtures; validators and policy gates; no-network loop dry run; hydrology proof-bearing thin slice; governed API; MapLibre shell integration; governed AI / Focus Mode; and domain lane expansion. This overlay supplies execution detail but does not override BLD-GREEN when phase sequencing conflicts. [IMPL-PIPE §25; BLD-GREEN §22]

Reconciliation Note - phase families: BLD-GREEN controls the greenfield phase sequence and v1.1 milestone additions; BLD-COMP controls PR-sized dependency mechanics; IMPL-PIPE controls loop-specific sequencing; IMPL-REF’s Phase A-H implementation plan is retained as lineage and NEEDS VERIFICATION because it depends on public-repo observations not freshly confirmed in this session. [BLD-GREEN §§22-23; BLD-COMP §4; IMPL-PIPE §25; IMPL-REF Implementation Plan]

4.5 20. Acceptance criteria [sources: BLD-GREEN §24; BLD-COMP §§3.2, 20, 30; IMPL-PIPE §§22, 26]

CONFIRMED / PROPOSED acceptance rule: KFM does not work merely because a folder tree, map layer, route, or model response exists. It works when the trust spine can show source admission, evidence resolution, policy decision, validation, release state, UI trust payload, correction path, and rollback target for at least one public-safe proof slice. [BLD-COMP §§1.2, 3, 30]

Acceptance area	Good-enough criterion	Status
Pre-RAW event family	Event schemas and valid/invalid fixtures exist; no-silent-RAW-admission test passes.	PROPOSED
RunReceipt and signing	RunReceipt schema, DSSE wrapper, verify script, stable spec_hash, offline fixture sign/verify.	PROPOSED
PromotionReceipt	Gates A-G produce joined decision records; PromotionReceipt is auditable record.	PROPOSED

Acceptance area	Good-enough criterion	Status
Merkle manifest	Proof pack includes Merkle field; leaf-mismatch test fails.	PROPOSED
OPA bootstrap	Access, obligations, gates, and release policy packages have positive/negative fixtures.	PROPOSED
KFM STAC profile	Required governance fields are enforced; nonconformant STAC item rejected.	PROPOSED
Catalog closure	PROV/STAC/DCAT emitters exist; catalog cannot precede proof-pack closure.	PROPOSED
NIST AI RMF alignment	AI governance mapping is documented; AI remains subordinate.	PROPOSED

PROPOSED negative acceptance: the system must demonstrate DENY, ABSTAIN, and ERROR paths, not only successful publication. A proof slice is incomplete if unknown rights are allowed, unresolved EvidenceRefs are answered, invalid fixtures pass, public routes read RAW/WORK/QUARANTINE, AI output bypasses citation validation, or release candidates lack rollback targets. [BLD-COMP §§5.2-5.3, 30.2; IMPL-PIPE §§22, 26]

4.6 21. Decision register (ADR index) [sources: BLD-GREEN §25; DOC-DIR §§2.4-2.5; BLD-COMP §5.3; IMPL-PIPE §28]

CONFIRMED ADR discipline: DOC-DIR requires ADRs before adding, removing, or renaming canonical roots; promoting compatibility roots; changing the schema-home rule; splitting or merging lifecycle phases; creating parallel homes for schemas, contracts, policy, sources, registries, releases, proofs, or receipts; or bending core invariants. [DOC-DIR §2.4]

Decision ID	Decision	Status
D-013	Use RunReceipt + DSSE/cosign for lifecycle receipt signing.	PROPOSED
D-014	Use Promotion Gates A-G as runnable workflow with decision_id join key.	PROPOSED
D-015	Add Merkle manifest layer to proof pack.	PROPOSED
D-016	Introduce pre-RAW event family before RAW.	PROPOSED
D-017	Use GitOps PR-emission for watcher-triggered promotions.	PROPOSED
D-018	Align AI boundary plan to NIST AI RMF Govern/Map/Measure/Manage.	PROPOSED documentation

CONFIRMED / PROPOSED control-loop ADR: IMPL-PIPE proposes ADR-0011 for the query-save-recompile loop and requires that loop outputs remain candidate deltas, receipts, validation reports, RecompileManifests, and promotion decisions until review and release gates pass. This is not autonomous self-modification and not a publication bypass. [IMPL-PIPE §§1, 26, 28]

4.7 22. Risk register [sources: BLD-GREEN §26; BLD-COMP §§3.3, 22, 27-31; IMPL-PIPE §§24, 27]

CONFIRMED risk posture: the dominant build risks are false implementation confidence, authority collision, public bypass, source-rights ambiguity, sensitive release, receipt/proof/catalog collapse, and derivative artifacts being mis-

taken for truth. BLD-GREEN v1.1 adds signing, OPA, Merkle, watcher, and AI replay risks. [BLD-GREEN §26; BLD-COMP §§3.3, 31; IMPL-PIPE §24]

Risk	Severity	Default outcome
Signing key exposure	Critical	DENY promotion; rotate key; re-sign affected receipts.
cosign / OPA version drift	High	ERROR; pin and rebuild.
Merkle root tampering	High	DENY; require independent manifest verification.
Silent watcher commit to main	High	DENY merge; require PR re-opening.
OPA default-allow on unknown input	High	DENY use; rebuild policy package with default deny.
Replay drift in AI prefilter	Medium	ERROR; rerun with pinned model and steward review.
Public bypass of RAW/WORK/QUARANTINE	High	DENY route, tile, or UI binding.
Sensitive prompt / source persistence	High	Persist summaries and hashes only; no chain-of-thought or secrets.

PROPOSED mitigation discipline: dependency admission records, no-network tests, path-role scans, secret scans, fixture drift checks, policy reason-code tests, finite-envelope tests, rollback drills, and public-negative-state UI tests should be added early. [BLD-COMP §§22, 27-31; IMPL-PIPE §§22-24]

4.8 23. Unknowns and verification backlog [sources: BLD-GREEN §27; BLD-COMP Appendix B; IMPL-PIPE §27; DOC-P18 §§11-14]

UNKNOWN current implementation state: repo topology, branch/dirty state, package manager, schema home, frameworks, policy engine, CI workflow, source rights, source terms, runtime logs, branch protections, source registries, emitted artifacts, deployment topology, dashboards, release artifacts, and current UI/API behavior remain unverified until a real checkout and runtime evidence are inspected. [IMPL-PIPE §27; BLD-COMP §2.1]

Backlog item	Default posture	Verification step
Current repo topology	UNKNOWN	Mount checkout; inspect roots and drift against DOC-DIR.
Schema-home convention	NEEDS VERIFICATION	Inspect ADR-0001 or equivalent; resolve drift by ADR.
Package manager / frameworks	UNKNOWN	Inspect lockfiles, app roots, workflow commands.
Policy engine	NEEDS VERIFICATION	Confirm OPA / Conftest / alternative and version pins.
Source rights and terms	NEEDS VERIFICATION	SourceActivationDecision before connector activation.
CI signing support	NEEDS VERIFICATION	Test cosign keyless or keyed flow in target CI.
OCI / S3 receipt storage	NEEDS VERIFICATION	Run ORAS or storage proof in controlled CI.
Watcher material-property allowlists	NEEDS VERIFICATION	Steward-review first 30 days of false-trigger behavior.
DSSE/cosign legal constraints	NEEDS VERIFICATION	Legal/steward review before regulated-source publication.
UI/API route behavior	UNKNOWN	Inspect code, tests, emitted envelopes, and logs.

PROPOSED verification order: begin with repo inspection and evidence inventory; then resolve schema-home/path ADRs; then run no-network schema, fixture, policy, and validator tests; then produce a dry-run PromotionReceipt, ReleaseManifest, ProofPack, RollbackCard, and Evidence Drawer payload for one public-safe hydrology claim/layer; then only after that admit live connectors, public UI breadth, or AI runtime providers. [IMPL-PIPE §§25-27; BLD-COMP Final build recommendation; BLD-GREEN §§21-24]

5 Part IV - Operations and Governance [sources: BLD-GREEN §§8, 16-20, 24-26; BLD-COMP §§5, 8-10, 20-23, 27-31; IMPL-PIPE §§17, 21-24, 27; DOC-DIR §§2.4-2.5, 4, 7.4; DOC-P18 §§8.6-8.10, 11-13; UIAI-GAI §§2-4; UIAI-MAP §§13-15; UIAI-MASTER §§10-14; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§12-13; DOM-FLORA §§11-12; DOM-HAZ §§1-2]

5.1 24. Validators and policy gates [sources: BLD-GREEN §§8, 16, 19, 24; BLD-COMP §§5, 8-10, 20, 30; IMPL-PIPE §17; DOC-DIR §§2.4, 4, 7.4; DOC-P18 §§8.6-8.7]

CONFIRMED doctrine: KFM validation is a promotion boundary, not a cleanup pass. Validators must prove that source admission, machine shape, object meaning, evidence closure, policy decision, sensitivity posture, release posture, correction path, and rollback support are adequate before public or semi-public exposure. BLD-GREEN v1.1 expands the suite around pre-RAW event schemas, RunReceipt, PromotionReceipt, Merkle manifests, OPA/Conftest policy-as-code, KFM STAC profile enforcement, catalog closure, and release integrity; those are CONFIRMED as doctrine and PROPOSED-to-create implementation unless a mounted repo later verifies them. [BLD-GREEN §§16, 24; BLD-COMP §§5, 20; IMPL-PIPE §17]

CONFIRMED placement rule: validator, schema, fixture, policy, and proof homes must follow responsibility-root placement. `schemas/` defines machine shape, `contracts/` defines meaning, `policy/` decides allow / deny / restrict / abstain, `tests/` proves enforcement, `fixtures/` holds sample cases, and `tools/` carries repo-wide validators or generators. Any change to schema-home, policy-home, lifecycle phases, or parallel homes requires an ADR. [DOC-DIR §§2.4-2.5, 4, 7.4]

PROPOSED gate architecture: validation should be layered so cheap deterministic checks run before evidence, policy, catalog, signing, or UI checks. A safe order is structural schema validation, fixture validation, source-role validation, EvidenceRef -> EvidenceBundle resolution, policy and sensitivity decision, lifecycle-state validation, receipt/proof validation, catalog-closure validation, release-manifest validation, and public-surface validation. [BLD-GREEN §§16, 19, 24; BLD-COMP §§5, 20, 30; IMPL-PIPE §17]

Gate family	Must answer	Default failure
Shape gate	Does the object match its schema and required version?	ERROR / quarantine
Meaning gate	Does it conform to contract and vocabulary?	ERROR / review
Source gate	Is source role, rights, cadence, and sensitivity known?	DENY / quarantine
Evidence gate	Do EvidenceRefs resolve to EvidenceBundles?	ABSTAIN
Policy gate	Is exposure allowed for this user, purpose, release class?	DENY
Lifecycle gate	Is object in correct RAW -> PUBLISHED state?	DENY
Receipt gate	Are RunReceipt / PromotionReceipt / decision logs present?	ERROR
Release gate	Does manifest include proof, correction, rollback?	DENY

CONFIRMED negative-state rule: validators must test DENY, ABSTAIN, ERROR, quarantine, stale, restricted, and review-needed paths, not only successful publication. [BLD-COMP §§20, 30; BLD-GREEN §24; IMPL-PIPE §§17, 22]

Reconciliation Note - policy tooling: BLD-GREEN v1.1 names OPA/Conftest as a proposed bootstrap, while DOC-DIR governs the policy home and IMPL-PIPE treats policy engine details as UNKNOWN without repo evidence. PROPOSED resolution: use OPA/Conftest as first policy-as-code candidate only after repo, dependency pins, and

CI environment are inspected; keep the PolicyDecision contract stable if the engine changes. [BLD-GREEN §§16-17; DOC-DIR §§2.4, 4; IMPL-PIPE §§17, 27]

5.2 25. CI/CD and delivery [sources: BLD-GREEN §§17, 22-24; BLD-COMP §§4-5, 27-31; IMPL-PIPE §23; UIAI-MASTER §12; IMPL-REF Connector Inventory]

UNKNOWN current CI/CD state: the supplied corpus does not verify current branch protections, workflow files, CI runner type, package manager, test commands, signing implementation, release storage, dashboards, or deployment posture. IMPL-REF reports earlier public-repo baseline checks and runtime surface, but BLD-COMP and IMPL-PIPE preserve that as lineage needing fresh verification. [IMPL-REF Connector Inventory; BLD-COMP §§2.1, 5.3; IMPL-PIPE §§23, 27]

PROPOSED CI sequence: the first pipeline should be no-network by default and should run path-role checks, schema validation, fixture validation, validator tests, policy tests, evidence-resolution tests, finite-envelope tests, promotion dry-run, release-deny tests, catalog-closure tests, and rollback-reference tests. Live connector tests, external endpoint checks, package-version checks, and runtime smoke tests should be separate, opt-in, source-activated jobs until rights, rate limits, secrets, and terms are verified. [BLD-GREEN §§17, 24; BLD-COMP §§5.3, 27-31; IMPL-PIPE §23]

PROPOSED delivery model: promotion should happen through visible, reviewed state transitions, not direct commits or artifact copies. BLD-GREEN v1.1 adds GitOps watcher / PR-emission as a proposed pattern for material changes; watchers create reviewable candidate changes and never publish directly. [BLD-GREEN §§17, 22-23; IMPL-PIPE §§21, 23; BLD-COMP §§8-10]

CI/CD lane	Purpose	Status
Path and drift scan	Enforce Directory Rules and ADR discipline	PROPOSED
Schema and fixture suite	Prove valid and invalid object shapes	PROPOSED
Validator suite	Prove lifecycle, evidence, policy, and release rules	PROPOSED
Policy suite	Prove deny-by-default and reason codes	PROPOSED
Promotion dry-run	Produce auditable gate outputs without publication	PROPOSED
Catalog closure	Emit or validate PROV / STAC / DCAT-style closure	PROPOSED
Signing / integrity	Verify RunReceipt, spec_hash, digest, Merkle patterns	PROPOSED / NEEDS VERIFICATION
UI / API smoke	Prove governed envelopes and no raw public paths	PROPOSED

PROPOSED release discipline: CI should block public release when unresolved evidence, unknown rights, unknown sensitivity, missing SourceActivationDecision, missing proof pack, invalid catalog closure, stale source state, missing rollback target, missing correction path, public RAW/WORK/QUARANTINE access, or direct model-client access is detected. [BLD-GREEN §§17, 19, 24; BLD-COMP §§20-23; IMPL-PIPE §§21, 23-24]

5.3 26. Testing strategy [sources: BLD-GREEN §§16, 24; BLD-COMP §§3.2, 5.3, 20, 30; IMPL-PIPE §22; DOC-P18 §§8.7, 11-13; domain fixture/test sections]

CONFIRMED doctrine / PROPOSED implementation: KFM testing must prove the trust spine, not only code execution. A passing suite must show that controlled inputs move through source admission, lifecycle state, validation, evidence resolution, policy decision, catalog/proof closure, release decision, governed API/UI payload, correction, and rollback without crossing forbidden boundaries. [BLD-GREEN §§16, 24; BLD-COMP §§3.2, 30; IMPL-PIPE §22]

PROPOSED test pyramid: start with deterministic no-network fixture tests, then schema and contract tests, validator unit tests, policy negative tests, evidence-resolution tests, lifecycle-state tests, receipt/proof tests, release-manifest tests, governed API envelope tests, UI trust-state tests, and only later live-source or runtime tests. [BLD-COMP §§5.3, 20; IMPL-PIPE §§22, 26; DOC-P18 §§8.7, 11-13]

PROPOSED fixture rule: every major object family should have at least one valid fixture, one invalid fixture, one denied fixture, one abstention fixture, and one rollback or correction fixture. Sensitive lanes should include public-safe transformed fixtures rather than real exact locations, living-person data, DNA material, rare-species coordinates, critical infrastructure details, or archaeological site geometry. [BLD-COMP §§20, 30; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§12-13; DOM-FLORA §§11-12]

Test class	Example assertion	Default status
Schema test	Required fields and versions are present	PROPOSED
Contract test	Object meaning matches vocabulary and lifecycle role	PROPOSED
Source-role test	A source is not used outside its authority	PROPOSED
Evidence test	EvidenceRef resolves or answer abstains	PROPOSED
Policy test	Unknown rights or sensitivity deny release	PROPOSED
Release test	Manifest has proof, correction, rollback	PROPOSED
UI trust test	Evidence Drawer shows state and negative outcomes	PROPOSED
AI boundary test	MockAdapter cannot answer without admissible evidence	PROPOSED

CONFIRMED non-regression posture: prior domain reports and IMPL-REF must be preserved as lineage, not silently overwritten. Domain expansion should include continuity maps, migration notes, alias/deprecation handling where needed, and non-regression tests before replacing prior scaffold semantics. [BLD-COMP §§2.1, 5.3; IMPL-REF Executive Summary; DOM-FAUNA §§1-4; DOM-HAB §§1-2]

5.4 27. Security and exposure posture [sources: BLD-GREEN §18; BLD-COMP §§21-23, 31; IMPL-PIPE §24; UIAI-MAP §13; UIAI-GAI §§2-4; UIAI-OLLAMA §§9-15; DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-HAZ §§1-2]

CONFIRMED doctrine: public and ordinary UI surfaces must deny direct access to RAW, WORK, QUARANTINE, unpublished candidate material, internal canonical stores, direct databases, direct source credentials, direct model clients, and exact sensitive geometry. Public surfaces consume released artifacts, governed API payloads, catalog records, tile services, EvidenceBundles, release manifests, and safe envelopes only. [BLD-GREEN §18; IMPL-PIPE §24; BLD-COMP §§21-23]

CONFIRMED AI exposure boundary: AI runtime access must sit behind governed API controls, evidence resolution, policy checks, citation validation, finite response envelopes, and receipts. UIAI-GAI recommends MockAdapter first; UIAI-OLLAMA permits Ollama only as a replaceable local or private runtime behind the governed boundary, not as a public truth surface or direct client endpoint. [UIAI-GAI §§2-4; UIAI-OLLAMA §§1, 9-15; IMPL-PIPE §20]

PROPOSED exposure controls: local deployments should use least privilege, deny-by-default CORS and network posture, secret isolation, audit logs, role-aware policy checks, no-public-admin shortcuts, no browser access to source credentials, and explicit review-console separation from public client behavior. [BLD-GREEN §18; BLD-COMP §§21-23, 31; UIAI-MAP §13]

CONFIRMED sensitivity posture: exact archaeology locations, living-person and DNA-derived material, rare-species occurrence geometry, culturally sensitive route or site information, critical infrastructure details, and high-risk hazards context require fail-closed exposure controls. [DOM-ARCH §§2-6; DOM-PEOPLE §§2-4; DOM-FAUNA §§12-13; DOM-FLORA §§11-12; DOM-HAZ §§1-2]

NEEDS VERIFICATION operational facts: package CVEs, dependency licenses, host hardening, model runtime settings, reverse-proxy rules, branch protections, signing key custody, storage bucket policy, source credentials, SSO/role mapping, audit retention, and backup/restore behavior are not established by the supplied corpus. [BLD-GREEN §§18, 26-27; UIAI-MAP §§13-15; IMPL-PIPE §27]

5.5 28. Catalog, proof, release, and publication [sources: BLD-GREEN §§11, 19, 24; BLD-COMP §§20-22; IMPL-PIPE §21; DOC-P18 §8.10; UIAI-MASTER §§10-12; domain catalog/proof sections]

CONFIRMED doctrine: catalog, proof, release, publication, and graph/triplet outputs are separate trust functions. The catalog supports discovery and closure; proof objects support claim and artifact validation; release manifests define the public artifact set and rollback target; publication is a governed state transition; graph, tile, PMTiles, COG, STAC/DCAT/PROV, UI, and AI outputs remain downstream carriers. [BLD-GREEN §§11, 19; IMPL-PIPE §21; UIAI-MASTER §§10-12]

PROPOSED catalog-closure pattern: a release candidate should not reach PUBLISHED until catalog records, provenance records, artifact manifests, evidence references, source roles, policy decisions, digests, review state, correction path, and rollback target agree. BLD-GREEN v1.1 proposes KFM STAC profile v1 and catalog closure as gated derived emission using PROV + STAC + DCAT; those remain proposed until implemented and verified. [BLD-GREEN §§11, 19, 24; DOC-P18 §8.10; UIAI-MASTER §§10-12]

PROPOSED proof-pack content: the smallest proof pack should include relevant SourceDescriptor or SourceActivationDecision, EvidenceBundle, validation report, policy decision, RunReceipt, PromotionReceipt or gate record, artifact digests, release manifest, correction lineage, rollback target, and public-surface payload references. This proof pack should allow a reviewer to understand why a claim or layer was allowed, denied, restricted, or abstained. [BLD-GREEN §§7, 19, 24; BLD-COMP §§20-22; IMPL-PIPE §21]

Publication object	Role	Must not become
Catalog record	Discovery and metadata closure	Proof by itself
Triples / graph edge	Derived relationship projection	Canonical truth
Proof pack	Evidence and validation support	Public UI substitute
ReleaseManifest	Published artifact set and rollback target	File-copy log
RunReceipt	Process memory and reproducibility record	Policy decision
PromotionReceipt	Auditable gate decision record	Human review replacement
Tile / PMTiles / COG	Renderable downstream carrier	Source of truth
AIRceipt	Runtime accountability record	EvidenceBundle replacement

5.6 29. Correction and rollback model [sources: BLD-GREEN §20; BLD-COMP §§21-23, 30-31; IMPL-PIPE §§21, 27; DOC-P18 §8.10; UIAI-MASTER §§10-14; domain rollback/correction sections]

CONFIRMED doctrine: correction and rollback are publication requirements, not afterthoughts. KFM's durable public unit is the inspectable claim; therefore a released claim, layer, catalog record, artifact, or answer must have a visible correction path and rollback target before it is treated as safely publishable. [BLD-GREEN §20; BLD-COMP §§21-22; IMPL-PIPE §21]

PROPOSED correction flow: a correction should preserve the original release record, identify the defect, classify the defect as evidence, source-role, rights, sensitivity, geometry, temporal, policy, validation, rendering, API, or AI-output related, emit a correction notice or equivalent record, update the relevant EvidenceBundle and release manifest, and publish a superseding release rather than silently mutating the old one. [BLD-GREEN §20; IMPL-PIPE §§21, 27; BLD-COMP §§21-23]

PROPOSED rollback flow: a rollback should identify the affected release, locate the prior safe artifact set, verify digests and manifests, disable or withdraw affected public surfaces, preserve audit receipts, mark stale or withdrawn UI state, and restore or republish the rollback target through the same governed release path. Rollback must not be a hidden file copy. [BLD-GREEN §20; UIAI-MASTER §§10-14; BLD-COMP §§21-23, 30-31]

Defect class	Correction posture	Rollback posture
Evidence gap	ABSTAIN or withdraw unsupported claim	Restore prior evidence-supported release

Defect class	Correction posture	Rollback posture
Rights defect	DENY public use; quarantine source/artifact	Withdraw affected artifacts
Sensitivity leak	Redact/generalize and notify stewards	Immediate public disablement
Geometry defect	Rebuild derivative layer and evidence payload	Restore previous digest-pinned artifact
Temporal defect	Correct valid/source/retrieval/release time	Mark stale until rebuilt
Policy defect	Re-run policy and decision envelope	Disable route/layer if gate failed
AI answer defect	Invalidate AIRceipt and response envelope	Remove answer; preserve EvidenceBundle
Catalog defect	Re-emit catalog closure after proof repair	Restore previous catalog state

CONFIRMED loop boundary: IMPL-PIPE's query-save-validate-compile-review-promote-recompile loop can help rebuild derived docs, indexes, catalogs, manifests, or proof packs, but it cannot publish or approve itself. Loop outputs remain candidate deltas, receipts, validation reports, and promotion decisions until review and release gates pass. [IMPL-PIPE §§1, 16, 21, 27]

6 Part V - Domain Lanes [sources: all DOM-* reports; DOM-HF; BLD-GREEN §§6, 21; IMPL-PIPE §14; DOC-ENC §§5, 7; DOC-DIR §§0-5; UIAI-MAP §§7-12]

Part V applies the fixed lane template requested for the unified PDF. CONFIRMED doctrine from prior parts governs every lane: domain material lives inside responsibility roots rather than new root folders; public clients use governed APIs and released artifacts; AI remains interpretive; and sensitive or weakly supported material fails closed. [DOC-DIR §§0-5; BLD-GREEN §§6, 21; IMPL-PIPE §14]

6.1 30.1 Hydrology [sources: DOM-HYD §§2-4; BLD-GREEN §§6, 21; IMPL-PIPE §14; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes watersheds, HUC/WBD units, stream networks, water observations, waterbody/flowline identity, terrain-derived hydrology context, and flood-regulatory context. NFHL-style flood layers must remain regulatory context, not observed inundation. [DOM-HYD §§2-4]

Sources of authority: DOM-HYD is authoritative inside the hydrology lane; BLD-GREEN treats hydrology as the preferred early proof lane; IMPL-PIPE treats hydrology as the safest first proof-bearing slice. [DOM-HYD §§2-4; BLD-GREEN §21; IMPL-PIPE §§14, 26]

Sensitivity posture: Hydrology is generally public-suitable when released, but operational freshness, source role, flood-regulatory interpretation, and emergency-use boundaries remain NEEDS VERIFICATION before public reliance. [DOM-HYD §§2-4; DOM-HAZ §§1-2]

Object families: PROPOSED objects include watershed unit, hydrologic observation, stream reach / flowline relation, source descriptor, EvidenceBundle, layer manifest, proof pack, RunReceipt, ReleaseManifest, and RollbackCard. [DOM-HYD §§2-4; BLD-GREEN §§7, 19-21]

Pipelines: PROPOSED first pipeline is fixture-first and no-network: HUC/WBD fixture, identity crosswalk fixture, observation normalization fixture, EvidenceBundle closure, catalog/proof closure, and public-safe layer output. [DOM-HYD §§2-4; IMPL-PIPE §26]

Publication gates: Gates require source activation, evidence closure, source-role distinction, freshness marking, validation, policy decision, release manifest, correction path, and rollback target. [DOM-HYD §§2-4; BLD-GREEN §§19-20]

Open verification items: Current endpoint behavior, source terms, package/tooling, schema home, emitted receipts, hydrology API routes, MapLibre layer binding, and CI execution remain UNKNOWN without mounted-repo evidence. [DOM-HYD §§2-4; IMPL-PIPE §27]

6.2 30.2 Soil [sources: DOM-SOIL §§1-3, 8-16; DOM-AG §§5-13; DOM-HYD §§2-4; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes soil map units, components, horizons, properties, pedon/profile context, soil moisture context, and governed adjacency to agriculture and hydrology. [DOM-SOIL §§8-16]

Sources of authority: DOM-SOIL governs soil-specific source roles and object families; DOM-AG and DOM-HYD govern adjacent agriculture/water relations without replacing soil authority. [DOM-SOIL §§8-16; DOM-AG §§5-13; DOM-HYD §§2-4]

Sensitivity posture: Soil products are usually public-safe at appropriate scale, but farm-specific, owner-specific, unpublished, proprietary, or operational sensor data requires rights and sensitivity review before release. [DOM-SOIL §§13-16; DOM-AG §§10-14]

Object families: PROPOSED objects include soil map unit, component, horizon, soil property observation, pedon/profile, soil moisture observation, suitability derivative, EvidenceBundle, catalog record, and ReleaseManifest. [DOM-SOIL §§8-16]

Pipelines: PROPOSED first slice is a static SSURGO/SDA-style snapshot plus one normalized soil-moisture sample, with validators, receipts, catalog stub, and public-safe derived layer. [DOM-SOIL §§9-16]

Publication gates: Gates require source-role checks, unit/parameter normalization, scale/fitness-for-use labeling, agriculture/hydrology relation checks, evidence closure, and release/rollback support. [DOM-SOIL §§13-16]

Open verification items: Current source terms, snapshot cadence, source endpoints, schema home, actual validator language, CI workflow, and repo implementation depth remain UNKNOWN / NEEDS VERIFICATION. [DOM-SOIL §§3, 20-26]

6.3 30.3 Habitat [sources: DOM-HAB §§1-2; DOM-HF §§1-5; DOM-FAUNA §§1-3; DOM-FLORA §§1-4; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes habitat patches, suitability surfaces, connectivity, restoration context, habitat assignment, and public-safe habitat products. [DOM-HAB §§1-2; DOM-HF §§1-5]

Sources of authority: DOM-HAB governs habitat lane architecture; DOM-HF governs the habitat-fauna thin-slice proof pattern; DOM-FAUNA and DOM-FLORA govern species-related boundaries. [DOM-HAB §§1-2; DOM-HF §§1-5]

Sensitivity posture: Habitat layers may reveal sensitive species context when joined to occurrence records; exact occurrence-linked habitat outputs must be generalized, redacted, reviewed, or denied when they create exposure risk. [DOM-HAB §§1-2; DOM-HF §§1-5; DOM-FAUNA §§12-13]

Object families: PROPOSED objects include habitat patch, habitat class, suitability model, connectivity relation, restoration candidate, habitat assignment, EvidenceBundle, DecisionEnvelope, and layer manifest. [DOM-HAB §§1-2; DOM-HF §§4-8]

Pipelines: PROPOSED first proof is a controlled habitat + fauna assignment using public-safe fixtures, not live sensitive source connectors. [DOM-HF §§1-5]

Publication gates: Gates require habitat evidence closure, species-sensitivity policy, model/derivative labeling, public-safe transformation receipt, release manifest, and rollback target. [DOM-HAB §§1-2; DOM-HF §§8-12]

Open verification items: Existing habitat files, schema-home convention, source rights, model fitness, live data permissions, and MapLibre/Evidence Drawer implementation remain UNKNOWN. [DOM-HAB §§1-2]

6.4 30.4 Fauna [sources: DOM-FAUNA §§1-3, 12-13; DOM-HF §§1-5; DOM-HAB §§1-2; DOC-P18 §8.6]

Scope: CONFIRMED / PROPOSED lane scope includes taxonomic identity, legal/conservation status, occurrence evidence, monitoring evidence, range/seasonal range, habitat relations, corridors, assemblages, invasive species, disease/mortality evidence, and public-safe derivatives. [DOM-FAUNA §§1-3]

Sources of authority: DOM-FAUNA governs fauna; DOM-HF governs the first habitat-assignment proof slice; habitat outputs remain adjacent derivatives rather than fauna truth. [DOM-FAUNA §§1-3; DOM-HF §§1-5]

Sensitivity posture: Sensitive taxa, nests, dens, roosts, hibernacula, spawning sites, steward-controlled records, and exact occurrence geometry fail closed unless a documented geoprivacy transform and review state allow release. [DOM-FAUNA §§12-13]

Object families: PROPOSED objects include taxon, occurrence, monitoring event, range, seasonal range, sensitivity classification, geoprivacy transform receipt, habitat assignment, EvidenceBundle, and ReleaseManifest. [DOM-FAUNA §§8-14]

Pipelines: PROPOSED first PR is synthetic and non-live-source, with source registry skeleton, public-safety validators, synthetic fixtures, and no activation of live wildlife connectors. [DOM-FAUNA §§1-3]

Publication gates: Gates require source-role registry, rights review, taxonomic resolution, sensitivity/geoprivacy policy, evidence closure, public-safe derivative check, and rollback support. [DOM-FAUNA §§9-13, 21-23]

Open verification items: Live connector rights, source roles, steward permissions, schema home, taxonomic resolver, exact geoprivacy policy, and current repo implementation remain UNKNOWN / NEEDS VERIFICATION. [DOM-FAUNA §§1-3]

6.5 30.5 Flora [sources: DOM-FLORA §§1-4, 11-12; DOM-HAB §§1-2; DOM-FAUNA §§1-2; DOC-P18 §8.6]

Scope: CONFIRMED / PROPOSED lane scope includes plant taxonomy, specimen and observation evidence, rare-plant controls, invasive plants, phenology, restoration context, vegetation surfaces, and public-safe flora products. [DOM-FLORA §§1-4]

Sources of authority: DOM-FLORA governs flora; habitat/fauna documents govern cross-lane joins without weakening flora-specific sensitivity and source-role discipline. [DOM-FLORA §§1-4; DOM-HAB §§1-2]

Sensitivity posture: Rare plants, steward-controlled records, exact sensitive occurrence geometry, and culturally sensitive plant knowledge require fail-closed treatment, redaction, generalization, review, or denial. [DOM-FLORA §§11-12]

Object families: PROPOSED objects include taxon, specimen, observation, occurrence, rare-plant sensitivity classification, invasive status, phenology observation, restoration candidate, EvidenceBundle, and layer manifest. [DOM-FLORA §§4, 10-15]

Pipelines: PROPOSED first slice is schema/fixture/validator-first and uses public-safe flora fixtures before any live source activation. [DOM-FLORA §§17-19]

Publication gates: Gates require taxonomic resolution, source role, rights review, rare-location policy, public-safe transform receipt, catalog/proof closure, release manifest, correction path, and rollback target. [DOM-FLORA §§11-17, 22-24]

Open verification items: Source endpoints, rights, steward review process, schema home, policy engine, MapLibre binding, proof-pack conventions, and current implementation remain UNKNOWN. [DOM-FLORA §§1-2, 23]

6.6 30.6 Hazards [sources: DOM-HAZ §§1-3; DOM-HYD §§2-4; DOM-AIR §§2-4; DOM-ROADS §§1-4; DOC-P18 §§8.3, 8.6]

Scope: CONFIRMED / PROPOSED lane scope includes hazard history, analysis, regulatory context, operational context, resilience review, map rendering, Evidence Drawer explanation, and governed AI summaries. [DOM-HAZ §§1-3]

Sources of authority: DOM-HAZ governs hazards; hydrology, atmosphere/air, roads/rail, and settlements lanes provide adjacent context without turning hazards into life-safety alerting. [DOM-HAZ §§1-3; DOM-HYD §§2-4; DOM-AIR §§2-4]

Sensitivity posture: Hazards outputs must not become emergency instructions or emergency alert systems; operational warnings require issue time, expiry time, source, retrieval time, freshness state, and official-source direction. [DOM-HAZ §§1-2]

Object families: PROPOSED objects include hazard event, warning/advisory context, disaster declaration, regulatory hazard area, detection, modeled derivative, resilience summary, EvidenceBundle, PolicyDecision, and Release-Manifest. [DOM-HAZ §§12-18]

Pipelines: PROPOSED pipelines keep historical events, operational warnings, regulatory areas, observations, detections, and models distinct through source-role taxonomy and freshness gates. [DOM-HAZ §§12-18]

Publication gates: Gates require evidence closure, source-role distinction, freshness/expiry handling, emergency-disclaimer posture, public-safe policy, release manifest, and rollback/correction support. [DOM-HAZ §§18-19]

Open verification items: Current official endpoints, licenses, feed schemas, rate limits, life-safety disclaimers, policy engine, and mounted implementation remain NEEDS VERIFICATION / UNKNOWN. [DOM-HAZ §§2, 8, 20]

6.7 30.7 Archaeology [sources: DOM-ARCH §§2-6; REF-3DGIS contents / Chs. 1-4; DOC-DIR §§0-3; DOC-P18 §§8.6, 8.9]

Scope: CONFIRMED / PROPOSED lane scope includes archaeological sites, records, surveys, artifacts, features, collections context, remote-sensing candidates, geophysical or LiDAR anomalies, 3D field documentation, review surfaces, and public-safe derived outputs. [DOM-ARCH §§2-6; REF-3DGIS Chs. 1-4]

Sources of authority: DOM-ARCH governs lane policy and implementation posture; REF-3DGIS is background for 3D archaeological practice and does not override KFM sensitivity rules. [DOM-ARCH §§2-6; REF-3DGIS Chs. 1-4]

Sensitivity posture: Exact archaeological site locations are denied by default; burial, human remains, sacred sites, cultural sensitivity, private-landowner privacy, collection security, and looting risk fail closed. [DOM-ARCH §§2-6]

Object families: PROPOSED objects include site assertion, archaeological feature, artifact/context record, survey record, remote-sensing candidate, sensitivity classification, steward review, transform receipt, EvidenceBundle, and ReleaseManifest. [DOM-ARCH §§9-23]

Pipelines: PROPOSED pipelines separate candidate detection from confirmed sites; LiDAR, aerial, satellite, geo-physical, model, or 3D anomalies remain candidates until source evidence and review support promotion. [DOM-ARCH §§2-6; REF-3DGIS Chs. 2-4]

Publication gates: Gates require source rights, cultural/steward review, sensitivity denial/default, generalized/suppressed geometry, transform receipts, catalog/proof closure, release manifest, correction path, and rollback target. [DOM-ARCH §§17-24]

Open verification items: Live source rights, steward permissions, exact-location rules, controlled access, schema home, policy tooling, and existing archaeology repo surfaces remain UNKNOWN / NEEDS VERIFICATION. [DOM-ARCH §§4-6]

6.8 30.8 Roads / Rail / Trade Routes [sources: DOM-ROADS §§1-4; DOM-SETTLE §§2-4; DOM-ARCH §§2-6; DOC-P18 §§8.11-8.13]

Scope: CONFIRMED / PROPOSED lane scope includes modern roads, historic roads, wagon roads, military/mail/emigrant/stage/cattle/trade corridors, rail corridors, depots, yards, crossings, restrictions, facilities, freight/logistics context, and graph projections. [DOM-ROADS §§1-4]

Sources of authority: DOM-ROADS governs transport-lane semantics; settlements/infrastructure and archaeology govern adjacent facilities, sensitive cultural routes, and site exposure constraints. [DOM-ROADS §§1-4; DOM-SETTLE §§2-4; DOM-ARCH §§2-6]

Sensitivity posture: Indigenous trade and mobility corridors, culturally sensitive routes, uncertain historic alignments, and critical facilities require steward review, uncertainty labeling, and public geometry generalization where precision would be harmful or unsupported. [DOM-ROADS §§1-4]

Object families: PROPOSED objects include road segment, rail segment, route, corridor, crossing, depot/facility, restriction event, operator/status event, graph edge, EvidenceBundle, and ReleaseManifest. [DOM-ROADS §§9-18]

Pipelines: PROPOSED pipelines normalize modern and historic transport evidence separately, create derived graph projections, and prevent routing/traversal graphs from replacing canonical records. [DOM-ROADS §§14-18]

Publication gates: Gates require source-role checks, temporal status checks, uncertainty/generalization, cultural-route sensitivity review, evidence closure, graph-derived labeling, release manifest, and rollback target. [DOM-ROADS §§13-19]

Open verification items: Current source rights, route precision, transport API/UI surfaces, graph implementation, schema home, CI, and deployed behavior remain UNKNOWN / NEEDS VERIFICATION. [DOM-ROADS §§2-4, 22]

6.9 30.9 Settlements / Infrastructure [sources: DOM-SETTLE §§2-4; DOM-ROADS §§1-4; REF-URBAN Chs. 1, 12-18; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes settlements, legal municipalities, census places, historic townsites, ghost towns, forts, depots, missions, reservation communities, infrastructure assets, networks, nodes, segments, facilities, service areas, operators, condition observations, dependencies, and public-safe representations. [DOM-SETTLE §§2-4]

Sources of authority: DOM-SETTLE governs settlements/infrastructure; roads/rail governs transport overlap; REF-URBAN provides planning/resilience background but not KFM publication authority. [DOM-SETTLE §§2-4; DOM-ROADS §§1-4; REF-URBAN Chs. 1, 12-18]

Sensitivity posture: Critical infrastructure, bridge/condition/inspection details, private or security-sensitive assets, and exact vulnerable facility geometry require policy gates, generalization, staged access, or denial. [DOM-SETTLE §§2-4, 18]

Object families: PROPOSED objects include settlement, legal place, historic place, infrastructure asset, network, node, segment, facility, service area, operator, condition observation, dependency relation, EvidenceBundle, and ReleaseManifest. [DOM-SETTLE §§7-15]

Pipelines: PROPOSED first slice is schema-and-fixture-first, with source descriptors, deterministic identity, validators, deny policies, no-network fixtures, and proof-pack/promotion fixtures before live source activation. [DOM-SETTLE §§2-4, 19]

Publication gates: Gates require legal/source-role distinction, sensitive infrastructure review, source rights, ge-

ometry generalization where needed, catalog/proof closure, release manifest, correction path, and rollback target. [DOM-SETTLE §§12-21]

Open verification items: Kansas source rights, exact infrastructure exposure rules, schema convention, app/API framework, proof storage, release process, and current repo files remain UNKNOWN / NEEDS VERIFICATION. [DOM-SETTLE §§3, 22]

6.10 30.10 Atmosphere / Air [sources: DOM-AIR §§2-4; DOM-HAZ §§1-3; UIAI-MAP §§8-10; DOC-P18 §§8.3, 8.12]

Scope: CONFIRMED / PROPOSED lane scope includes air observations, AQI reports, climate context, smoke, Earth-observation layers, model fields, anomaly surfaces, fusion products, and public-safe air-context layers. [DOM-AIR §§2-4]

Sources of authority: DOM-AIR governs air/atmosphere knowledge-character discipline; hazards governs operational-context boundaries; UIAI-MAP governs map display as downstream rendering. [DOM-AIR §§2-4; DOM-HAZ §§1-3; UIAI-MAP §§8-10]

Sensitivity posture: Observations, public AQI reports, regulatory archives, models, smoke masks, anomaly surfaces, and fusion products are not epistemically interchangeable; public output must label knowledge character and freshness. [DOM-AIR §§2-4]

Object families: PROPOSED objects include station observation, parameter registry item, AQI summary, climate normal/context, smoke mask, model field, EO-derived layer, anomaly surface, EvidenceBundle, and layer manifest. [DOM-AIR §§2-4, 12-18]

Pipelines: PROPOSED first PR is docs/registry/schema/fixture/validator/policy/dry-run only, with no live fetch, no public promotion, and no UI/API binding beyond typed contract notes. [DOM-AIR §§2-4]

Publication gates: Gates require source role, parameter/unit validation, knowledge-character labels, freshness state, evidence closure, policy decision, release manifest, and rollback/correction support. [DOM-AIR §§2-4, 15-23]

Open verification items: Current live rights, terms, quotas, source schemas, attribution requirements, source cadence, and mounted implementation depth remain UNKNOWN / NEEDS VERIFICATION. [DOM-AIR §§2-4]

6.11 30.11 Geology / Natural Resources [sources: DOM-GEOL §§1-4, 7-18; DOM-SOIL §§8-16; DOM-HYD §§2-4; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes bedrock geology, surficial geology, stratigraphy, lithology, geologic age, structures, geomorphology, hydrostratigraphic context, boreholes, well logs, cores, measured sections, geophysics, geochemistry, mineral occurrences, resource deposits, extraction/reclamation context, and public-safe generalized layers. [DOM-GEOL §§1-4, 7-18]

Sources of authority: DOM-GEOL governs geology/resource semantics; soil and hydrology provide adjacent relation context only. [DOM-GEOL §§1-4; DOM-SOIL §§8-16; DOM-HYD §§2-4]

Sensitivity posture: Geologic and resource layers may be public-safe when generalized, but resource estimates, extraction sites, borehole details, private/proprietary data, and sensitive geometry require rights and exposure review. [DOM-GEOL §§8, 16]

Object families: PROPOSED objects include geologic unit, lithology, age/correlation, structure, borehole/well-log/core reference, geochemistry/geophysics record, resource occurrence, resource estimate, public-safe geometry, EvidenceBundle, and ReleaseManifest. [DOM-GEOL §§7-18]

Pipelines: PROPOSED first slice starts with source descriptors, schema index, fixtures, source-role policy, public-safe geometry checks, catalog closure, and one offline fixture slice. [DOM-GEOL §§1-4, 26]

Publication gates: Gates require geology/resource anti-collapse checks, source-role validation, public-safe geometry, rights review, evidence closure, catalog/proof/release separation, and rollback path. [DOM-GEOL §§8, 16-23]

Open verification items: Actual source rights, schema authority, package manager, validator language, database/graph stack, CI, emitted proofs, and runtime behavior remain UNKNOWN. [DOM-GEOL §§3-5, 25]

6.12 30.12 Agriculture [sources: DOM-AG §§1-5, 9-14; DOM-SOIL §§8-16; DOM-HYD §§2-4; REF-URBAN planning/resilience chapters; DOC-ENC §§5, 7]

Scope: CONFIRMED / PROPOSED lane scope includes crop observations, field/land-use context, agricultural management signals, irrigation/water adjacency, soil adjacency, stress indicators, production context, public-safe agricultural layers, and governed indicators. [DOM-AG §§5-14]

Sources of authority: DOM-AG governs agriculture; DOM-SOIL and DOM-HYD govern adjacent soil/water claims; planning references provide background only. [DOM-AG §§1-5; DOM-SOIL §§8-16; DOM-HYD §§2-4]

Sensitivity posture: Agriculture requires rights and privacy review for farm-specific, owner-specific, proprietary, operational, or fine-grained management data; grid or indicator outputs must not be equated with private ground truth. [DOM-AG §§5-14]

Object families: PROPOSED objects include crop observation, agricultural unit / public-safe field representation, management observation, irrigation relation, soil relation, water relation, stress indicator, source descriptor, EvidenceBundle, and ReleaseManifest. [DOM-AG §§5-14]

Pipelines: PROPOSED expansion preserves prior scaffold semantics through living documentation, source registries, contracts, validators, policies, tests, CI, catalog closure, proof/receipt separation, API/UI trust payloads, migration notes, and rollback controls. [DOM-AG §§1-3, 6-14]

Publication gates: Gates require source rights, source-role boundaries, public-safe geometry, farm/privacy sensitivity, indicator uncertainty, evidence closure, catalog/proof closure, release manifest, correction path, and rollback target. [DOM-AG §§12-19]

Open verification items: Actual source terms, package manager, schema home, OPA/Conftest/Cosign pins, API/UI framework, proof storage, CODEOWNERS, and existing agriculture files remain UNKNOWN / NEEDS VERIFICATION. [DOM-AG §§1-3, 20]

6.13 30.13 People / Genealogy / DNA / Land [sources: DOM-PEOPLE §§2-4; DOC-DIR §§0-3; BLD-GREEN §§18-20; REF-TEMPDB Chs. 2, 10-11; REF-DDD definitions / bounded context]

Scope: CONFIRMED / PROPOSED lane scope includes person assertions, relationship hypotheses, genealogy evidence, DNA-derived material, land instruments, temporal ownership assertions, parcel/geometry relations, and public-safe historical research outputs. [DOM-PEOPLE §§2-4]

Sources of authority: DOM-PEOPLE governs people/genealogy/DNA/land; REF-TEMPDB supports temporal assertion modeling as background; REF-DDD supports bounded-context language as background. [DOM-PEOPLE §§2-4; REF-TEMPDB Chs. 2, 10-11; REF-DDD definitions]

Sensitivity posture: Living-person and DNA-derived outputs are denied or restricted by default; relationship hypotheses remain hypotheses; assessor/tax records are not title truth; parcel geometry is not title-boundary proof without source role and evidence. [DOM-PEOPLE §§2-4]

Object families: PROPOSED objects include person assertion, person record, relationship assertion, DNA evidence restriction object, land instrument, ownership assertion, parcel relation, temporal validity record, EvidenceBundle, PolicyDecision, and ReleaseManifest. [DOM-PEOPLE §§2-4]

Pipelines: PROPOSED pipelines keep source documents, GEDCOMs, DNA match files, land instruments, assessor rolls, and geometry sources in governed lifecycle states and prevent public artifacts from reading RAW, WORK, or QUARANTINE directly. [DOM-PEOPLE §4; DOC-DIR §§0-4]

Publication gates: Gates require living-person screening, DNA restriction, source terms, evidence closure, temporal validity, title/assessor/geometry source-role distinction, sensitivity policy, review state, release manifest, correction path, and rollback target. [DOM-PEOPLE §§2-4; BLD-GREEN §§18-20]

Open verification items: Existing people/genealogy/DNA/land surfaces, source rights, steward permissions, privacy policy, schema convention, CI gates, public access classes, and current repo implementation remain UNKNOWN / NEEDS VERIFICATION. [DOM-PEOPLE §§2-4]

7 Part VI - Closing [sources: BLD-GREEN §§28-29; BLD-COMP §§30-32, Appendices A-C; IMPL-PIPE §§29-30, Appendices A-E; DOC-ENC §§15-16; DOC-P18 §§10-15; DOC-DIR §§0-5; REF-DDD definitions; REF-TEMPDB Chs. 2, 10-11; REF-API Chs. 1-4; REF-GISPRIMER Parts I-II]

CONFIRMED closing posture: Part VI consolidates final readiness checklist, shared vocabulary, and appendix plan. It does not add new doctrine, does not claim repo implementation, and does not bypass Phase 4 consolidation. [BLD-GREEN §§28-29; IMPL-PIPE §§29-30; BLD-COMP Appendices A-C]

7.1 31. Final build checklist [sources: BLD-GREEN §28; BLD-COMP §§30-32, Appendix A; IMPL-PIPE §30; DOC-ENC §§15-16; DOC-P18 §§11-15]

CONFIRMED / PROPOSED checklist rule: KFM is not ready because it has a map, an app, a model, a repo tree, or a generated PDF; it is ready only when a small public-safe slice can prove source admission, evidence closure, policy decision, validation, release, public surface behavior, correction, and rollback. [BLD-GREEN §28; BLD-COMP §§30-32; IMPL-PIPE §30]

Checklist item	Release-blocking question	Truth posture
Source ledger and duplicates	Are duplicate PDFs, superseded docs, lineage docs, and reference docs separated from active authority?	CONFIRMED doctrine / PROPOSED execution
Directory Rules conformance	Does each proposed file belong to correct responsibility root, with ADRs for root/schema-home changes?	CONFIRMED doctrine
Lifecycle integrity	Does the slice preserve RAW -> WORK/QUARANTINE -> PROCESSED -> CATALOG/TRIPLET -> PUBLISHED?	CONFIRMED doctrine
Schema / contract / fixture baseline	Are object meaning, machine shape, valid fixtures, invalid fixtures, and tests present before public use?	PROPOSED execution
Source activation	Are source role, rights, cadence, steward, sensitivity, and release class known before connector activation?	PROPOSED execution
Evidence closure	Does each consequential claim resolve EvidenceRef -> EvidenceBundle or abstain?	CONFIRMED doctrine
Policy and sensitivity	Are DENY, ABSTAIN, ERROR, restricted, stale, and review-needed states tested, not hidden?	CONFIRMED doctrine / PROPOSED execution
Receipts and promotion	Are RunReceipt, PromotionReceipt or equivalent gate records, validation reports, and logs emitted?	PROPOSED execution
Catalog/proof/release separation	Are catalog, proof pack, release manifest, receipts, corrections, and rollback targets distinct?	CONFIRMED doctrine
Public API/UI boundary	Do public clients avoid RAW, WORK, QUARANTINE, candidates, direct stores, and direct model clients?	CONFIRMED doctrine

Checklist item	Release-blocking question	Truth posture
Domain sensitivity gates	Are rare species, archaeology, infrastructure, living-person, DNA, cultural, and sovereignty risks fail-closed?	CONFIRMED doctrine
Correction and rollback	Can reviewers identify prior safe release, affected artifacts, correction notice, and rollback path?	PROPOSED execution

PROPOSED readiness determination: the first acceptance target should be one no-network proof slice, preferably hydrology or a habitat/fauna synthetic fixture, because those lanes are repeatedly presented as safer early proof lanes than living-person, DNA, exact archaeology, rare-species, or sensitive infrastructure lanes. [BLD-GREEN §§21, 28; BLD-COMP §§30-32; IMPL-PIPE §§26, 30; DOM-HYD §§2-4; DOM-HF §§1-5]

UNKNOWN implementation readiness: current repo topology, schema files, tests, CI workflows, branch protections, emitted receipts, public routes, runtime logs, dashboards, source terms, package pins, deployment settings, and actual UI behavior remain unverified in the supplied corpus. [IMPL-PIPE §§27, 30; BLD-COMP §2.1; DOC-P18 §11]

7.2 32. Glossary [sources: IMPL-PIPE §29; BLD-GREEN Appendix E; BLD-COMP §32; DOC-DIR §§0-4; DOC-ENC §§3-4; REF-DDD definitions; REF-TEMPDB Chs. 2, 10-11; REF-API Chs. 1-4; REF-GISPRIMER Parts I-II]

CONFIRMED glossary rule: KFM object-family names are preserved as stable vocabulary, while external references such as DDD, temporal SQL, API design, and GIS/cartography supply background language only. External-reference terms do not outrank KFM doctrine, Directory Rules, evidence posture, or publication gates. [IMPL-PIPE §29; BLD-GREEN Appendix E; DOC-DIR §§0-4; REF-DDD definitions]

7.2.1 32.1 Trust and evidence terms [sources: IMPL-PIPE §29; BLD-GREEN §§7, 10-11; BLD-COMP §§3, 6, 10; UIAI-GAI §§2-4]

Term	Unified definition
Inspectable claim	Public or semi-public statement reconstructable to evidence, scope, source role, policy posture, review state, release state, and correction lineage.
EvidenceRef	Pointer from a claim, feature, answer, layer, or proof item to evidence that must resolve before consequential release.
EvidenceBundle	Resolved evidence package supporting a claim, including source, scope, provenance, policy, citation, and review context.
SourceDescriptor	Machine-readable source identity, source role, rights, cadence, access, steward, sensitivity, and release posture.
SourceActivationDecision	Gate record deciding whether a source may be used, restricted, quarantined, denied, or held for review.
PolicyDecision	Explicit allow, deny, restrict, abstain, or error decision tied to user, purpose, release class, evidence, and sensitivity.
DecisionEnvelope	Finite decision wrapper used by APIs, runtime surfaces, and UI/AI payloads to avoid ambiguous truth states.
RuntimeResponseEnvelope	Governed AI/API response wrapper carrying outcome, evidence context, citations, policy state, and validation result.
AIRceipt	Runtime accountability record for bounded AI use; never a substitute for EvidenceBundle.

Term	Unified definition
RunReceipt	Execution record pinning inputs, outputs, hashes, tool versions, timestamps, failures, policy posture, and evidence refs.
PromotionReceipt	Governed state-transition record enumerating Promotion Gates A-G as auditable promotion memory.
ReleaseManifest	Record of published artifact set, digests, policy posture, release state, correction path, and rollback target.

7.2.2 32.2 Lifecycle and repository terms [sources: DOC-DIR §§0-5; BLD-GREEN front matter, Appendix E; IMPL-PIPE §§7, 29; BLD-COMP §§4-5]

Term	Unified definition
RAW	Admitted source material under source identity; not a public surface.
WORK	Transformation and candidate space for material not ready for processed or public use.
QUARANTINE	Governed holding state for rights, sensitivity, validation, source-role, evidence, temporal, or policy defects.
PROCESSED	Normalized outputs that have passed transformation checks but are not automatically public.
CATALOG / TRIPLET	Discovery, claim, relationship, graph, and publication-support surfaces downstream of evidence and processing.
PUBLISHED	Released, policy-allowed, reviewable, rollback-capable public or semi-public artifact state.
Responsibility root	Repo root whose name encodes a governance responsibility such as docs, schemas, policy, tests, data, release, apps, or tools.
Compatibility root	Existing or source-reported root retained for compatibility until ADR/migration confirms authority, deprecation, or drift resolution.
Domain lane	Domain-specific architecture and workflow inside responsibility roots; not a new root folder by topic name.
Trust membrane	Boundary ensuring public/UI/AI surfaces consume governed APIs and released artifacts, not raw or internal stores.
Pre-RAW event family	event_envelope, prefilter_output, and event_run_receipt schemas governing attempted admission before RAW.
Watcher	Pipeline that detects material changes and opens a PR rather than committing directly or publishing directly.

7.2.3 32.3 Release, integrity, and loop terms [sources: BLD-GREEN Appendix E; IMPL-PIPE §§16, 29; DOC-P18 §§8.5, 8.10; UIAI-MASTER §§10-12]

Term	Unified definition
DSSE	Dead Simple Signing Envelope; wrapper used to carry a canonicalized receipt and signature.
Merkle manifest	Tamper-evident tree of leaf hashes over the canonical release file set.

Term	Unified definition
spec_hash	Deterministic hash of a canonicalized spec or receipt body used to join CI, policy, attestation, and publication records.
STAC KFM profile v1	PROPOSED STAC extension adding required governance properties for evidence, bundle, release, policy, and correction state.
CandidateDelta	Proposed loop-generated change that cannot promote until validation and review pass.
QueryRunRecord	Record of a governed query/save iteration and its admissible evidence context.
EvidenceResolutionRecord	Record showing how EvidenceRefs were resolved, denied, or left unresolved.
RecompileManifest	Record of deterministic inputs, compiler/spec hashes, rebuilt artifacts, output hashes, and rollback target.
LoopDecision	Decision about whether a loop candidate proceeds, abstains, is denied, errors, or requires review.
LoopRollbackReference	Link from a loop-generated artifact to the prior safe state or release target.
PromotionDecision	Recorded decision that moves a candidate across a governed state transition.
Correction lineage	Chain showing defect, correction, supersession, affected artifacts, evidence change, and public notice path.

7.2.4 32.4 External-reference terms used as background [sources: REF-DDD definitions; REF-TEMPDB Chs. 2, 10-11; REF-API Chs. 1-4; REF-GISPRIMER Parts I-II; REF-3DGIS Chs. 1-4]

Term	KFM use
Bounded context	Used to keep domain-lane language coherent; does not override Directory Rules or KFM root placement.
Ubiquitous language	Used for consistent domain vocabulary within a lane and across contracts, schemas, tests, and UI labels.
Valid time	Time when a fact or condition is true in the modeled world; distinct from retrieval, release, and correction time.
Transaction time	Time when KFM records or changes knowledge; useful for audit and correction lineage.
Bitemporal modeling	Modeling both valid time and transaction time where claims, land assertions, observations, or releases require it.
API contract	Agreement about public resource behavior and payload shape; in KFM, always governed by evidence and policy boundaries.
Geographic representation	Simplified communication of place, scale, projection, geometry, and meaning; not neutral truth by itself.
3D GIS artifact	Spatial visualization or analysis carrier for archaeological or terrain context; not confirmation without evidence and review.

7.3 33. Appendices [sources: Phase 1 ledger; Phase 2 crosswalk; Phase 4 consolidation diff; BLD-GREEN §29 and Appendices A-E; IMPL-PIPE Appendices A-E; BLD-COMP Appendices A-C; DOC-P18 §§3, 10-15; DOC-ENC §3]

CONFIRMED appendix rule: appendices preserve traceability without inflating duplicate or superseded material into independent authority. [BLD-GREEN §29; IMPL-PIPE Appendices A-E; BLD-COMP Appendices A-C]

7.3.1 33.1 Appendix A - unified-section to source-doc crosswalk [sources: Phase 2 crosswalk; all listed sources]

Unified section(s)	Primary source-doc sections	Notes
0.1-0.5	BLD-GREEN front matter; IMPL-PIPE front matter; BLD-COMP front matter; DOC-DIR §0; DOC-ENC §3; DOC-P18 §3	Title, truth labels, ledger, reading rule, ToC.
1-5	BLD-GREEN §§1-2, 8, 18-20; IMPL-PIPE §§1, 4, 7, 17, 21, 24; DOC-DIR §§0-5; BLD-COMP §§1-3	Doctrine and boundaries; Directory Rules wins placement conflicts.
6-8	BLD-GREEN §§7, 10-11; IMPL-PIPE §§6-8, 15-16, 21; BLD-COMP §§3, 6, 10; UIAI-GAI §§2-4	System overview, lifecycle, object families.
9-11	DOC-DIR §§2-5, §7.4; BLD-GREEN §§3-5, 9; IMPL-PIPE §§9-13; BLD-COMP §§4-8	Repository, schema/contract, source registry.
12	BLD-GREEN §6; IMPL-PIPE §14; DOC-ENC §§5, 7; all DOM-* reports	Domain lane architecture summary.
13-15	BLD-GREEN §§12-15; UIAI-MAP §§1, 4-15; UIAI-WHOLE §§17-25; UIAI-GAI §§1-4; UIAI-MASTER §§9-14	API, UI/MapLibre, governed AI.
16-23	BLD-GREEN §§21-28; BLD-COMP §§3-5, 20-31; IMPL-PIPE §§25-28; IMPL-REF Executive Summary	Build plan, decisions, risks, unknowns.
24-29	BLD-GREEN §§16-20, 24-26; BLD-COMP §§20-23, 27-31; IMPL-PIPE §§17, 21-24; DOC-P18 §§8.6-8.10	Operations, validation, CI/CD, testing, security, release, rollback.
30.1-30.13	DOM-HYD; DOM-SOIL; DOM-HAB; DOM-FAUNA; DOM-FLORA; DOM-HAZ; DOM-ARCH; DOM-ROADS; DOM-SETTLE; DOM-AIR; DOM-GEOL; DOM-AG; DOM-PEOPLE	Domain lanes.
31-33	BLD-GREEN §§28-29; BLD-COMP §§30-32, Appendices A-C; IMPL-PIPE §§29-30, Appendices A-E; DOC-P18 §§10-15	Closing checklist, glossary, appendices.

7.3.2 33.2 Appendix B - superseded, duplicate, lineage, and reference handling [sources: Phase 1 ledger; BLD-GREEN front matter; IMPL-PIPE front matter; DOC-DIR §0; IMPL-REF Executive Summary; UIAI-MASTER §§3-4; DOC-P18 §3]

Item	Handling in unified PDF	Reason
BLD-GREEN-DUP	Duplicate of BLD-GREEN; not independent authority.	Same v1.1 build plan content by duplicate upload.
IMPL-REF-DUP	Duplicate of IMPL-REF; not second implementation source.	Same implementation-reference content by duplicate upload.
IMPL-PIPE-DUP	Duplicate of IMPL-PIPE; not second pipeline authority.	Same v0.3 pipeline manual content by duplicate upload.
BLD-GREEN v1.0	Superseded by BLD-GREEN v1.1; retained as lineage.	v1.1 states it supersedes v1.0 and is additive.
IMPL-PIPE v0.2	Retained as lineage; v0.3 controls loop doctrine.	v0.3 states v0.2 is retained as baseline/lineage.
Prior Directory_Rules.pdf	Superseded where DOC-DIR conflicts.	DOC-DIR §0 states it supersedes prior initial form.
IMPL-REF repo claims	Retained as NEEDS VERIFICATION lineage.	Current mounted repo evidence is absent.
Prior domain scaffolds	Preserve as lineage; do not treat as current implementation.	Domain PDFs are PDF-only outputs or no-mounted-repo plans.
UIAI-MASTER v1.8	Retained through v1.9 appendix lineage.	v1.9 states prior baseline appended and retained.
Pass packets	Treat as idea inventory and expansion backlog unless promoted.	DOC-P18 separates ledger, weak material, gaps, and agenda.
External textbooks	Use as REF background only.	They are not KFM doctrine, repo evidence, or publication authority.

7.3.3 33.3 Appendix C - consolidation diff summary [sources: Phase 4 consolidation diff; BLD-COMP §§2.3, 5.3; DOC-P18 §§10-13]

CONFIRMED: Phase 4 records editorial consolidation decisions made across Parts I-VI. PROPOSED: these changes become final PDF content through this Phase 5 assembly; no implementation state is upgraded by consolidation. [Phase 4; BLD-GREEN front matter; BLD-COMP §2.3; IMPL-PIPE §1]

Diff family	Consolidation action	Result
Source authority	Deduplicated duplicate uploads and downgraded external textbooks to REF/background.	Duplicate files remain ledgered but not independent authority.
Implementation state	Reconciled BLD-GREEN zero-start assumption with IMPL-REF public-repo claims.	Current implementation maturity remains UNKNOWN / NEEDS VERIFICATION.
Core invariant	Preserved once in §2 and cross-referenced later.	Avoids repeated doctrine while keeping invariant visible.
Truth posture	Preserved truth-label definitions and separated system outcomes.	DENY/ABSTAIN/ERROR are outcomes, not truth labels.
Directory Rules	Made DOC-DIR controlling for path/responsibility roots.	Domain reports cannot create root folders by topic.
Object families	Consolidated shared objects in §8 and moved domain-only objects to Part V.	Prevents cross-domain object duplication.
Schema-home conflict	Resolved only as PROPOSED default to DOC-DIR <code>schemas/contracts/v1/<...></code> rule.	ADR or repo inspection still required.
UI/AI conflict	Reconciled adapter-first governed AI with Ollama runtime guidance.	Contracts/gates/receipts first; providers later.

Diff family	Consolidation action	Result
Build phases	BLD-GREEN controls phase sequence; BLD-COMP supplies PR mechanics; IMPL-PIPE supplies loop overlay.	No invented phase names beyond source evidence.
Operations	Consolidated validators, CI, testing, security, publication, and rollback into gate families.	Negative states remain test requirements.
Domain lanes	Compressed detailed domain reports into fixed lane templates.	Domain-specific sensitivity preserved without repeating global doctrine.
Glossary	Grouped KFM terms and external-reference terms.	Object-family names preserved verbatim.

7.3.4 33.4 Appendix D - open items carried forward [sources: BLD-GREEN §27; IMPL-PIPE §27; BLD-COMP Appendix B; DOC-P18 §§11-13]

Carry-forward item	Status	Reason not resolved
Current repo topology and implementation maturity	UNKNOWN	No mounted current repo, tests, workflows, logs, manifests, dashboards, or release artifacts were inspected.
Schema-home implementation	NEEDS VERIFICATION	DOC-DIR gives a default; IMPL-REF reports unresolved ambiguity; current repo evidence is absent.
Actual API routes and DTOs	UNKNOWN	UI/API reports are implementation plans, not current route inventories.
Policy engine and CI tools	NEEDS VERIFICATION	OPA/Conftest/cosign/DSSE are proposed patterns, not verified installed tooling.
Source rights and live connectors	NEEDS VERIFICATION	Domain reports require terms, rights, cadence, steward, and sensitivity review before activation.
Public-sensitive domain release readiness	NEEDS VERIFICATION	Rare species, archaeology, DNA/living-person, infrastructure, cultural/sovereignty, and hazards material require fail-closed gates.
MapLibre package/tool/version posture	NEEDS VERIFICATION	UIAI-MAP and UIAI-MASTER contain version-sensitive items requiring current checks.
AI runtime/provider posture	NEEDS VERIFICATION	Ollama or any provider is admissible only after adapter contracts, evidence gates, citation validation, and receipts.
Release/correction/rollback rehearsal	PROPOSED	The build plan requires proof through one no-network public-safe slice before broad release.

7.3.5 33.5 Appendix E - PDF assembly notes [sources: Phase 5 protocol; PDF skill render/verify workflow]

CONFIRMED Phase 5 assembly note: this PDF was generated as a single simple LaTeX build from the approved Phase 1-4 draft content and rendered for layout verification. No PDF binary is embedded in chat. [Phase 5 protocol]